

Agentic LLMs for AKI prediction

July 4, 2026

Research Goal: Find scholarly literature relevant to designing an agentic LLM framework—specifically focusing on LLMs using tool-calling to orchestrate external quantitative or time-series models mid-inference—for predicting drug-induced acute kidney injury. Include broader AKI and nephrotoxicity prediction papers when their modeling setup is transferable. The search should prioritize papers that inform this agentic multi-model orchestration as the core novelty, followed by papers that ground the clinical prediction task in high-density longitudinal EHR data (prioritizing ICU settings like MIMIC-IV, alongside general inpatient data). Finally, include papers on the generation of synthetic EHR data via differential privacy, strictly used as a privacy-preserving infrastructure for model training.

Found 133 papers · July 4, 2026 · Estimated coverage of relevant papers: 80%

Summary of Results

The literature is asymmetric: robust longitudinal AKI/nephrotoxicity predictors are abundant, and tool-using clinical LLM agents are now credible, but explicit AKI systems that *orchestrate external quantitative models mid-inference* remain scarce—task-specific evidence is mainly [1], while the clearest transferable agent patterns come from calculator-centered agents such as [3, 2, 7, 5].

Agentic orchestration patterns

- **Calculator/tool selection and execution** is the most mature paradigm for evidence-based risk prediction: AgentMD curates executable calculators and invokes them at inference [3, 2]; RiskAgent extends this to hundreds of validated tools [7, 5].
- **Important boundary:** many clinical multi-agent papers are not true tool-calling systems.
 - AKI-specific multi-agent work exists [1, 6], but often emphasizes deliberation/structure-following rather than external model invocation.
 - Transferable EHR-access infrastructure comes from code/tool agents over structured records [4, 17, 56].
 - Hybrid expert-model + LLM orchestration is exemplified by ColaCare [10], but usually summarizes model outputs rather than dynamically choosing among quantitative renal-risk models.

Clinical prediction substrate for drug-induced AKI

- **ICU longitudinal AKI prediction** is strongest in dense EHR settings: continuous/dynamic forecasting [26], multimodal MIMIC-IV models [8], KDIGO-aware time-series baselines [35], and multicenter/external validations [65, 67, 78].
- **Drug-induced/nephrotoxic AKI** is the most transferable subtask for agentic routing:
 - Dynamic inpatient nephrotoxin-exposed risk modeling [9], DRN time-series models across nephrotoxic drugs [21], adult NINJA-style recurrent models [12].
 - Drug-specific models for colistin [36], vancomycin [82, 48], cisplatin [100, 96], and aminoglycosides [59].

- Phenotype/causality framing is anchored by DIKD standardization [58] and ICU drug-AKI association studies [46].

Privacy-preserving training infrastructure

- **Formal DP synthetic longitudinal EHR** is best represented by heterogeneous/irregular generators such as IGAMT [37] and earlier DP sequence generators [85, 70].
- **High-fidelity non/weak-formal privacy baselines** for ICU time series include diffusion and mixed-type generators [13, 22, 23], useful as utility references when selecting a DP approach.

Paper Catalog (133 papers)

	Year	Cit/yr	Title	Authors	Journal
1	2024	0.4	AKI-Detector: A Multi-Agent Framework by Integrating Machine Learning and Large Language Models for Early Prediction of Acute Kidney Injury in ICU. (link)	Tongyue Shi, Meirong Xiao, Haowei Xu, Huiying Zhao, and Guilan Kong	AMIA ... Annual Symposium proceedings. AMIA Symposium
2	2025	5.8	AgentMD: Empowering language agents for risk prediction with large-scale clinical tool learning (link)	Qiao Jin et al.	Nature Communications
3	2024	18	AgentMD: Empowering Language Agents for Risk Prediction with Large-Scale Clinical Tool Learning (link)	Qiao Jin et al.	ArXiv
4	2024	39	EHRAgent: Code Empowers Large Language Models for Few-shot Complex Tabular Reasoning on Electronic Health Records (link)	Wenqi Shi et al.	Proceedings of the Conference on Empirical Methods in Natural Language Processing. Conference on Empirical Methods in Natural Language Processing
5	2025	4.9	RiskAgent: Autonomous Medical AI Copilot for Generalist Risk Prediction (link)	Fenglin Liu et al.	ArXiv
6	2025		AKIBoards: A Structure-Following Multiagent System for Predicting Acute Kidney Injury (link)	D. Gordon, P. Petousis, Susanne B. Nicholas, and Alex A. T. Bui	ArXiv
7	2025		RiskAgent: Synergizing Language Models with Validated Tools for Evidence-Based Risk Prediction (link)	Fenglin Liu et al.	

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8	2024	4.1	Forecasting acute kidney injury and resource utilization in ICU patients using longitudinal, multimodal models (link)	Yukun Tan et al.	Journal of biomedical informatics
9	2019	1.5	Development and validation of an automated algorithm for identifying patients at higher risk for drug-induced acute kidney injury. (link)	Nakyung Jeon, Benjamin J. Staley, Carl H. Henriksen, Gloria P. Lipori, and A. Winterstein	American journal of health-system pharmacy : AJHP : official journal of the American Society of Health-System Pharmacists
10	2024	21	ColaCare: Enhancing Electronic Health Record Modeling through Large Language Model-Driven Multi-Agent Collaboration (link)	Zixiang Wang et al.	Proceedings of the ACM on Web Conference 2025
11	2026	4.0	Vital Trace: Protocol-Constrained Patient-State Reasoning for Longitudinal Clinical Trajectories (link)	Zhan Qu and Michaela K. Farber	
12	2024	3.4	Predicting Nephrotoxic Acute Kidney Injury in Hospitalized Adults: A Machine Learning Algorithm (link)	Benjamin R. Griffin et al.	Kidney Medicine
13	2023	16	Reliable generation of privacy-preserving synthetic electronic health record time series via diffusion models (link)	Muhang Tian, Bernie Chen, Allan Guo, Shiyi Jiang, and Anru R. Zhang	Journal of the American Medical Informatics Association : JAMIA
14	2025	4.4	MoMA: a mixture-of-multimodal-agents architecture for enhancing clinical prediction modelling (link)	Jifan Gao et al.	NPJ Digital Medicine
15	2026		TreeText-CTS: Compact, Source-Traceable Tree-Path Evidence for Irregular Clinical Time-Series Prediction (link)	Kwanhyung Lee et al.	
16	2026		Beyond binary AKI classification: development and external validation of a distributional model predicting serum creatinine and urine output trajectories in ICU patients (link)	Jef Jonkers et al.	Critical Care
17	2025	2.5	EHR-MCP: Real-world Evaluation of Clinical Information Retrieval by Large Language Models via Model Context Protocol (link)	Kanato Masayoshi et al.	ArXiv

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18	2023	1251	Toolformer: Language Models Can Teach Themselves to Use Tools (link)	Timo Schick et al.	ArXiv
19	2026		Forecasting ICU Acute Kidney Injury with Actionable Lead Time Using Interpretable Machine Learning: Development and Multi-Center Validation (link)	A. Hourani, Z. Jakubowska, and J. Małyszko	Journal of Clinical Medicine
20	2024	0.8	Enhancing Acute Kidney Injury Prediction through Integration of Drug Features in Intensive Care Units (link)	Gabriel D. M. Manalu, Mulomba Mukendi Christian, Songhee You, and Hyebyong Choi	ArXiv
21	2023	2.1	Time Series AI Model for Acute Kidney Injury Detection Based on a Multicenter Distributed Research Network: Development and Verification Study (link)	S. Heo et al.	JMIR Medical Informatics
22	2021	28	Generating synthetic mixed-type longitudinal electronic health records for artificial intelligent applications (link)	Jin Li, B. Cairns, Jingsong Li, and T. Zhu	NPJ Digital Medicine
23	2023	30	EHR-Safe: generating high-fidelity and privacy-preserving synthetic electronic health records (link)	Jinsung Yoon et al.	NPJ Digital Medicine
24	2025	15	Artificial intelligence models for predicting acute kidney injury in the intensive care unit: a systematic review of modeling methods, data utilization, and clinical applicability (link)	Tongyue Shi, Yu Lin, Huiying Zhao, and Guilan Kong	JAMIA Open
25		7	EHRFlow : A Large Language Model-Driven Iterative Multi-Agent Electronic Health Record Data Analysis Workflow (link)	Hao Wu et al.	
26	2019	133	A Clinically Applicable Approach to Continuous Prediction of Future Acute Kidney Injury (link)	Nenad Tomašev et al.	Nature
27	2026		AKI-twinX: explainable organ structured digital twin for sepsis AKI trajectory forecasting (link)	Jinjin Cai et al.	medRxiv

	Year	Cit/yr	Title	Authors	Journal
28	2026		Machine learning studies of drug-induced nephrotoxicity: a scoping review (link)	Mawardi Ihsan, Shun-Ting Chang, Wei-Kai Chan, and Hsiang-Yin Chen	Therapeutic Advances in Drug Safety
29	2025		Predicting Acute Kidney Injury with Nephrotoxic Burden in Noncritical Patients (link)	Esra Adiyeké et al.	Kidney360
30	2026		D2MDT: Department-aware Multidisciplinary Team Consultation with Deliberation for Efficient Clinical Prediction (link)	Yongqi Liang et al.	
31	2026		Agentifying Patient Dynamics within LLMs through Interacting with Clinical World Model (link)	Minghao Wu et al.	
32	2026		Computer-Interpretable Domain Knowledge for Drug-Induced Acute Kidney Injury: a Knowledge Graph Approach. (link)	Romy Vos et al.	Scientific data
33	2022	50	TALM: Tool Augmented Language Models (link)	Aaron Parisi, Yao Zhao, and Noah Fiedel	ArXiv
34	2025	29	Large language model agents can use tools to perform clinical calculations (link)	Alex J. Goodell, Simon N. Chu, D. Rouholiman, and Larry F. Chu	NPJ Digital Medicine
35	2024	5.4	An empirical study on KDIGO-defined acute kidney injury prediction in the intensive care unit (link)	Xinrui Lyu et al.	Bioinformatics
36	2024	4.9	Machine Learning Algorithms to Predict Colistin-Induced Nephrotoxicity from Electronic Health Records in Patients with Multidrug-Resistant Gram-Negative Infection. (link)	Ling-Wan Chiu et al.	International journal of antimicrobial agents
37	2024	3.5	IGAMT: Privacy-Preserving Electronic Health Record Synthesization with Heterogeneity and Irregularity (link)	Wenjie Wang et al.	Proceedings of the ... AAAI Conference on Artificial Intelligence. AAAI Conference on Artificial Intelligence

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38	2025		DMAP: A Deep-Model Driven Multi-Agent Framework for Reliable Diagnosis Prediction (link)	Wenjing Yu et al.	2025 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)
39	2025	2.4	Advancing Continuous Prediction for Acute Kidney Injury via Multi-Task Learning: Towards Better Clinical Applicability (link)	Hyunwoo Kim et al.	IEEE Journal of Biomedical and Health Informatics
40	2026		ClinicalAgents: Multi-Agent Orchestration for Clinical Decision Making with Dual-Memory (link)	Zhuohan Ge et al.	
41	2017	16	Prediction of Acute Kidney Injury With a Machine Learning Algorithm Using Electronic Health Record Data (link)	Hamid Mohamdlou et al.	Canadian Journal of Kidney Health and Disease
42	2023	1.5	Redefining digital health interfaces with large language models (link)	F. Imrie, Paulius Rauba, and M. Schaar	Frontiers in Artificial Intelligence
43	2024	33	MedCalc-Bench: Evaluating Large Language Models for Medical Calculations (link)	Nikhil Khandekar et al.	ArXiv
44	2026	6.0	EHR-RAG: Bridging Long-Horizon Structured Electronic Health Records and Large Language Models via Enhanced Retrieval-Augmented Generation (link)	Lang Cao, Qingyu Chen, and Yue Guo	ArXiv
45	2023		A dual-channel deep learning approach to continuous prediction of acute kidney injury in the intensive care unit (link)	Xuetong Kong et al.	2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)
46	2023	5.7	Acute kidney injury associated with nephrotoxic drugs in critically ill patients: a multicenter cohort study using electronic health record data (link)	Izak A R Yasrebi-de Kom et al.	Clinical Kidney Journal
47	2019	21	A prospective multi-center quality improvement initiative (NINJA) indicates a reduction in nephrotoxic acute kidney injury in hospitalized children. (link)	S. Goldstein et al.	Kidney international

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48	2025	3.5	Development and Validation of a CatBoost-Based Model for Predicting Significant Creatinine Elevation in ICU Patients Receiving Vancomycin Therapy (link)	Junyi Fan et al.	BioMedInformatics
49	2017	106	Real-valued (Medical) Time Series Generation with Recurrent Conditional GANs (link)	Cristóbal Esteban, Stephanie L. Hyland, and G. Rätsch	ArXiv
50	2023		Causal prediction models for medication safety monitoring: The diagnosis of vancomycin-induced acute kidney injury (link)	Izak A. R. Yasrebi-de Kom et al.	ArXiv
51	2026	2.0	Privately Fine-Tuned LLMs Preserve Temporal Dynamics in Tabular Data (link)	Lucas Rosenblatt, Peihan Liu, Ryan McKenna, and N. Ponomareva	ArXiv
52	2025		EMR-AGENT: Automating Cohort and Feature Extraction from EMR Databases (link)	Kwanhyung Lee et al.	ArXiv
53	2026	2.0	Planner-Auditor Twin: Agentic Discharge Planning with FHIR-Based LLM Planning, Guideline Recall, Optional Caching and Self-Improvement (link)	Kaiyuan Wu, Aditya Nagori, and R. Kamaleswaran	ArXiv
54	2026		When OpenClaw Meets Hospital: Toward an Agentic Operating System for Dynamic Clinical Workflows (link)	Wenxian Yang et al.	
55	2025		Temporal Concept Tracing: Making Deep Learning Predictions Interpretable and Actionable for ICU Acute Kidney Injury Prevention (link)	S. S. I. Badhon et al.	Proceedings of the AAAI Symposium Series
56	2025	24	MedAgentBench: A Realistic Virtual EHR Environment to Benchmark Medical LLM Agents (link)	Yixing Jiang et al.	
57	2024	9.4	Prompting Large Language Models for Zero-Shot Clinical Prediction with Structured Longitudinal Electronic Health Record Data (link)	Yinghao Zhu et al.	ArXiv

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58	2015	15	Phenotype Standardization for Drug Induced Kidney Disease (link)	R. Mehta et al.	Kidney international
59	2025	8.1	Machine learning modeling for the risk of acute kidney injury in inpatients receiving amikacin and etimicin (link)	Peitao Zhang et al.	Frontiers in Pharmacology
60	2020	7.6	Validation of the usefulness of artificial neural networks for risk prediction of adverse drug reactions used for individual patients in clinical practice (link)	S. Imai et al.	PLoS ONE
61	2022	1.2	AKI-BERT: a pre-trained clinical language model for early prediction of acute kidney injury (link)	Chengsheng Mao, Liang Yao, and Yuan Luo	BMC Medical Informatics and Decision Making
62	2023	27	Making Language Models Better Tool Learners with Execution Feedback (link)	Shuofei Qiao, Honghao Gui, Huajun Chen, and Ningyu Zhang	ArXiv
63	2025	8.9	From Scores to Steps: Diagnosing and Improving LLM Performance in Evidence-Based Medical Calculations (link)	Ben Wang et al.	Proceedings of the Conference on Empirical Methods in Natural Language Processing. Conference on Empirical Methods in Natural Language Processing
64	2025	20	TIMER: temporal instruction modeling and evaluation for longitudinal clinical records (link)	Hejie Cui et al.	NPJ Digital Medicine
65	2022	10	Generalizability of an acute kidney injury prediction model across health systems (link)	Jie Cao et al.	Nature machine intelligence
66	2023	1.8	A Machine Learning Algorithm Predicting Acute Kidney Injury in Intensive Care Unit Patients (NAVOY Acute Kidney Injury): Proof-of-Concept Study (link)	Inger Persson, Adam Grünwald, Ludvine Morvan, David Becedas, and Martin Arlbrandt	JMIR Formative Research
67	2024	2.8	Development and validation of a deep interpretable network for continuous acute kidney injury prediction in critically ill patients (link)	Meicheng Yang et al.	Artificial intelligence in medicine

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68	2018	1.8	Electronic Health Record–Enabled Big-Data Approaches to Nephrotoxin-Associated Acute Kidney Injury Risk Prediction (link)	S. Sutherland	Pharmacotherapy: The Journal of Human Pharmacology and Drug Therapy
69	2025	2.6	Development and validation of machine learning models to predict vancomycin- and teicoplanin-associated acute kidney injury: a retrospective, multicenter study. (link)	Pinyi Lo, F. Chan, Yi-En Ku, San-Yuan Wang, and Hsiangyin Chen	International journal of antimicrobial agents
70	2020	28	Differentially Private Synthetic Medical Data Generation using Convolutional GANs (link)	A. Torfi, E. Fox, and Chandan K. Reddy	Inf. Sci.
71	2026		RAG-Augmented Explainable Early Warning System for Acute Kidney Injury in the ICU: A KDIGO-Aligned XGBoost Pipeline with SHAP-Driven Evidence Retrieval on MIMIC-IV (link)	Kerem Gencer	IEEE Access
72	2025	1.1	Scaling Clinician-Grade Feature Generation from Clinical Notes with Multi-Agent Language Models (link)	Jiayi Wang et al.	
73	2022	4.6	Development and validation of a deep neural network-based model to predict acute kidney injury following intravenous administration of iodinated contrast media in hospitalized patients with chronic kidney disease: a multicohort analysis. (link)	Ping Yan, S. Duan, Xiao Luo, Ning-Ya Zhang, and Ying-Hao Deng	Nephrology, dialysis, transplantation : official publication of the European Dialysis and Transplant Association - European Renal Association
74	2025	0.7	Decode Like a Clinician: Enhancing LLM Fine-Tuning with Temporal Structured Data Representation (link)	Daniel Fadlon et al.	IJCNLP-AACL
75	2025	3.4	Tool-wielding language model-based agent offers conversational exploration of clinical tabular data (link)	Andrew J Yang et al.	npj Artificial Intelligence
76	2026	2.0	MedMCP-Calc: Benchmarking LLMs for Realistic Medical Calculator Scenarios via MCP Integration (link)	Yakun Zhu et al.	ArXiv

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77	2026		FeatEHR-LLM: Leveraging Large Language Models for Feature Engineering in Electronic Health Records (link)	Hojjat Karami, David Atienza, J.-P. Thiran, and Anisoara Ionescu	
78	2024	4.6	Development and external validation of deep learning clinical prediction models using variable-length time series data (link)	Fereshteh S. Bashiri et al.	Journal of the American Medical Informatics Association : JAMIA
79	2021	9.2	Convolutional Neural Network Model for Intensive Care Unit Acute Kidney Injury Prediction (link)	S. Le et al.	Kidney International Reports
80	2021	8.2	Real-Time Clinical Decision Support Based on Recurrent Neural Networks for In-Hospital Acute Kidney Injury: External Validation and Model Interpretation (link)	Kipyo Kim et al.	Journal of Medical Internet Research
81	2013	19	Electronic Health Record Identification of Nephrotoxin Exposure and Associated Acute Kidney Injury (link)	S. Goldstein et al.	Pediatrics
82	2022	2.5	Analysis of a machine learning–based risk stratification scheme for acute kidney injury in vancomycin (link)	Fei Mu et al.	Frontiers in Pharmacology
83	2019	4.1	Utility of Electronic Medical Record Alerts to Prevent Drug Nephrotoxicity. (link)	Melissa L. Martin and F. P. Wilson	Clinical journal of the American Society of Nephrology : CJASN
84	2023	11	A randomized clinical trial assessing the effect of automated medication-targeted alerts on acute kidney injury outcomes (link)	F. Wilson et al.	Nature Communications
85	2020	11	Generating sequential electronic health records using dual adversarial autoencoder (link)	Dongha Lee et al.	Journal of the American Medical Informatics Association : JAMIA
86	2020	1.2	Generation of Differentially Private Heterogeneous Electronic Health Records (link)	Kieran Chin-Cheong, T. Sutter, and Julia E. Vogt	ArXiv

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87	2023	0.7	Privacy-Preserving Data Synthesis via Differentially Private Normalizing Flows with Application to Electronic Health Records Data (link)	Bingyue Su, Yu Wang, Daniele E. Schiavazzi, and Fang Liu	Proceedings of the AAAI Symposium Series
88	2023	0.6	Differentially Private Normalizing Flows for Density Estimation, Data Synthesis, and Variational Inference with Application to Electronic Health Records (link)	Bingyue Su, Yu Wang, Daniele E. Schiavazzi, and Fang Liu	ArXiv
89	2017	51	Privacy-Preserving Generative Deep Neural Networks Support Clinical Data Sharing (link)	Brett K. Beaulieu-Jones et al.	Circulation. Cardiovascular quality and outcomes
90	2023	1.8	Synthesize high-dimensional longitudinal electronic health records via hierarchical autoregressive language model (link)	Brandon Theodorou, Cao Xiao, and Jimeng Sun	Nature Communications
91	2022	30	A Multifaceted benchmarking of synthetic electronic health record generation models (link)	Chao Yan et al.	Nature Communications
92	2020	13	SynTEG: a framework for temporal structured electronic health data simulation (link)	Ziqi Zhang, Chao Yan, T. Lasko, Jimeng Sun, and B. Malin	Journal of the American Medical Informatics Association : JAMIA
93	2020	11	EVA: Generating Longitudinal Electronic Health Records Using Conditional Variational Autoencoders (link)	Siddharth Biswal et al.	Machine Learning in Health Care
94	2017	5.4	Predicting Inpatient Acute Kidney Injury over Different Time Horizons: How Early and Accurate? (link)	Peng Cheng, L. Waitman, Yong Hu, and Mei Liu	AMIA ... Annual Symposium proceedings. AMIA Symposium
95	2024	2.6	Interpretable machine learning-based individual analysis of acute kidney injury in immune checkpoint inhibitor therapy (link)	Minoru Sakuragi et al.	PLOS ONE
96	2025	1.3	Prediction of Cisplatin-Induced Acute Kidney Injury Using an Interpretable Machine Learning Model and Electronic Medical Record Information (link)	Kaori Ambe et al.	Clinical and Translational Science

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97	2018	4.7	Multi-perspective predictive modeling for acute kidney injury in general hospital populations using electronic medical records (link)	Jianqin He et al.	JAMIA Open
98	2020	7.5	A prediction and interpretation framework of acute kidney injury in critical care (link)	K. Gong, H. Lee, Kaiye Yu, Xiaolei Xie, and Jingshan Li	Journal of biomedical informatics
99	2025	8.2	Enhancing Clinical Decision Support and EHR Insights through LLMs and the Model Context Protocol: An Open-Source MCP-FHIR Framework (link)	Abul Ehtesham, Aditi Singh, and Saket Kumar	2025 IEEE World AI IoT Congress (AllIoT)
100	2022	4.3	Prediction model of acute kidney injury induced by cisplatin in older adults using a machine learning algorithm (link)	Takaya Okawa et al.	PLoS ONE
101	2023	2.9	Detection of potential drug-drug interactions for risk of acute kidney injury: a population-based case-control study using interpretable machine-learning models (link)	Hayato Akimoto et al.	Frontiers in Pharmacology
102	2022	12	PromptEHR: Conditional Electronic Healthcare Records Generation with Prompt Learning (link)	Zifeng Wang and Jimeng Sun	Proceedings of the Conference on Empirical Methods in Natural Language Processing. Conference on Empirical Methods in Natural Language Processing
103	2022	1.1	Application of Causal Discovery Algorithms in Studying the Nephrotoxicity of Remdesivir Using Longitudinal Data from the EHR (link)	Jianqiu Zhang et al.	AMIA ... Annual Symposium proceedings. AMIA Symposium
104	2026		Risk Horizons: Structured Hypothesis Spaces for Longitudinal Clinical Prediction (link)	Zhan Qu and Michaela K. Farber	
105	2024	4.6	SepsisCalc: Integrating Clinical Calculators into Early Sepsis Prediction via Dynamic Temporal Graph Construction (link)	Changchang Yin et al.	KDD : proceedings. International Conference on Knowledge Discovery & Data Mining

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106	2021	21	Membership inference attacks against synthetic health data (link)	Ziqi Zhang, Chao Yan, and B. Malin	Journal of biomedical informatics
107	2022	12	Machine learning models for predicting acute kidney injury: a systematic review and critical appraisal (link)	I. Vagliano et al.	Clinical Kidney Journal
108	2026		ReallICU: Do LLM Agents Understand Long-Context ICU Data? A Benchmark Beyond Behavior Imitation (link)	Cheng-nan Shen et al.	
109	2015	1.2	Detection of Drug–Drug Interactions Inducing Acute Kidney Injury by Electronic Health Records Mining (link)	Y. Girardeau et al.	Drug Safety
110	2021	26	Benchmarking Differentially Private Synthetic Data Generation Algorithms (link)	Yuchao Tao, Ryan McKenna, Michael Hay, Ashwin Machanavajjhala, and G. Miklau	ArXiv
111	2023	13	Characterization of Risk Prediction Models for Acute Kidney Injury (link)	Yunlin Feng et al.	JAMA Network Open
112	2022	3.6	Keeping synthetic patients on track: feedback mechanisms to mitigate performance drift in longitudinal health data simulation (link)	Ziqi Zhang, Chao Yan, and B. Malin	Journal of the American Medical Informatics Association : JAMIA
113	2021	4.0	Multitask prediction of organ dysfunction in the intensive care unit using sequential subnetwork routing (link)	Subhrajit Roy et al.	Journal of the American Medical Informatics Association : JAMIA
114	2026		Large language model-augmented offline reinforcement learning framework for sepsis management in critical care. (link)	Yoo-Soon Lim et al.	NPJ digital medicine
115	2021	4.2	Prediction and visualization of acute kidney injury in intensive care unit using one-dimensional convolutional neural networks based on routinely collected data (link)	Noriaki Sato et al.	Computer methods and programs in biomedicine

	Year	Cit/yr	Title	Authors	Journal
116	2024	15	CEHR-GPT: Generating Electronic Health Records with Chronological Patient Timelines (link)	Chao Pang et al.	ArXiv
117			Safety-aware explainable deep reinforcement learning for nephrotoxic medication management in critical care (link)	Hyunwoo Kim et al.	Biomed. Signal Process. Control.
118	2025	4.5	No Black Boxes: Interpretable and Interactable Predictive Healthcare with Knowledge-Enhanced Agentic Causal Discovery (link)	Xiaoxue Han et al.	Conference on Empirical Methods in Natural Language Processing
119	2019	4.7	Differentially Private Mixed-Type Data Generation For Unsupervised Learning (link)	U. Tantipongpipat, Chris Waites, Digvijay Boob, Amaresh Ankit Siva, and Rachel Cummings	ArXiv
120	2024	4.3	SynEHRgy: Synthesizing Mixed-Type Structured Electronic Health Records using Decoder-Only Transformers (link)	Hojjat Karami, David Atienza, and Anisoara Ionescu	ArXiv
121	2025		Generating Multi-Table Time Series EHR from Latent Space with Minimal Preprocessing (link)	Eunbyeol Cho, Jiyoun Kim, Minjae Lee, Sungjin Park, and Edward Choi	ArXiv
122	2016		Use of EMRs to Identify Drugs that Induce Acute Kidney Injury (link)	B. Melton, N. Graham, Russ Waitman, and Mei Liu	Summit on Clinical Research Informatics
123	2022	5.4	Conditional generation of medical time series for extrapolation to underrepresented populations (link)	Simon Bing, Andrea Dittadi, Stefan Bauer, and P. Schwab	PLOS Digital Health
124	2019	39	DP-CGAN: Differentially Private Synthetic Data and Label Generation (link)	Reihaneh Torkzadehmahani, P. Kairouz, and B. Paten	2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)
125	2022	14	A Survey of Generative Adversarial Networks for Synthesizing Structured Electronic Health Records (link)	Ghadeer O. Ghosheh, Jin Li, and T. Zhu	ACM Computing Surveys

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126	2025	2.5	Term2Note: Synthesising Differentially Private Clinical Notes from Medical Terms (link)	Yuping Wu et al.	ArXiv
127	2025		EagleCare: Augmenting ICU Monitoring with LLMs and Knowledge Graph Reasoning (link)	Jiawei Guo et al.	2025 IEEE International Conference on Knowledge Graph (ICKG)
128	2024	1.7	Controllable Synthetic Clinical Note Generation with Privacy Guarantees (link)	Tal Baumel et al.	ArXiv
129	2021	3.9	Prediction Models for AKI in ICU: A Comparative Study (link)	Qing Qian, Jinming Wu, Jiayang Wang, Haixia Sun, and Lei Yang	International Journal of General Medicine
130	1986	2.3	A model for predicting nephrotoxicity in patients treated with aminoglycosides. (link)	Charles L. Sawyers, Richard D. Moore, Stephen A. Lerner, and Craig R. Smith	The Journal of infectious diseases
131	2026		Large Language Model Automated Extraction of Clinical Signs and Symptoms From Emergency Department Reports for Machine Learning Prediction Models: Development and Validation Study (link)	Anoeska Schipper et al.	JMIR Medical Informatics
132	2023	287	AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation Framework (link)	Qingyun Wu et al.	ArXiv
133	2024	125	Magentic-One: A Generalist Multi-Agent System for Solving Complex Tasks (link)	Adam Fourney et al.	ArXiv

Paper Details

1. · 100% match · 2024 · 0.4 cit/yr

AKI-Detector: A Multi-Agent Framework by Integrating Machine Learning and Large Language Models for Early Prediction of Acute Kidney Injury in ICU. ([link](#))

Tongyue Shi, Meirong Xiao, Haowei Xu, Huiying Zhao, and Guilan Kong

AMIA ... *Annual Symposium proceedings. AMIA Symposium* · 1 citations

2. · 100% match · 2025 · 5.8 cit/yr

AgentMD: Empowering language agents for risk prediction with large-scale clinical tool learning ([link](#))

Qiao Jin et al.

Nature Communications · Oct 23, 2025 · 4 citations

Clinical calculators play a vital role in healthcare, but their utilization is often hindered by usability and dissemination challenges. We introduce AgentMD, a novel language agent capable of curating and applying clinical calculators across various clinical contexts. As a tool builder, AgentMD first uses PubMed to curate a diverse set of 2,164 executable clinical calculators with over 85% accuracy for quality checks and over 90% pass rate for unit tests. As a tool user, AgentMD autonomously selects and applies the relevant clinical calculators. Our evaluations show that AgentMD significantly outperforms GPT-4 for risk prediction (87.7% vs. 40.9% in accuracy). Results on 698 real-world emergency department notes confirm that AgentMD accurately computes medical risks at an individual level. Moreover, AgentMD can provide population-level insights for institutional risk management. Our study illustrates the capabilities of language agents to curate and utilize clinical calculators for both individual patient care and at-scale healthcare analytics. Clinical calculators play a vital role in healthcare, but their utilization remains to be optimized. The authors present AgentMD, an AI agent that can autonomously curate clinical calculators from medical literature and apply them to various use cases.

3. · 100% match · 2024 · 18 cit/yr

AgentMD: Empowering Language Agents for Risk Prediction with Large-Scale Clinical Tool Learning ([link](#))

Qiao Jin et al.

ArXiv · Feb 20, 2024 · 43 citations

Clinical calculators play a vital role in healthcare by offering accurate evidence-based predictions for various purposes such as prognosis. Nevertheless, their widespread utilization is frequently hindered by usability challenges, poor dissemination, and restricted functionality. Augmenting large language models with extensive collections of clinical calculators presents an opportunity to overcome these obstacles and improve workflow efficiency, but the scalability of the manual curation process poses a significant challenge. In response, we introduce AgentMD, a novel language agent capable of curating and applying clinical calculators across various clinical contexts. Using the published literature, AgentMD has automatically curated a collection of 2,164 diverse clinical calculators with executable functions and structured documentation, collectively named RiskCalcs. Manual evaluations show that RiskCalcs tools achieve an accuracy of over 80% on three quality metrics. At inference time, AgentMD can automatically select and apply the relevant RiskCalcs tools given any patient description. On the newly established RiskQA benchmark, AgentMD significantly outperforms chain-of-thought prompting with GPT-4 (87.7% vs. 40.9% in accuracy). Additionally, we also applied AgentMD to real-world clinical notes for analyzing both population-level and risk-level patient characteristics. In summary, our study illustrates the utility of language agents augmented with clinical calculators for healthcare analytics and patient care.

4. · 100% match · 2024 · 39 cit/yr

EHRAgent: Code Empowers Large Language Models for Few-shot Complex Tabular Reasoning

on Electronic Health Records ([link](#))

Wenqi Shi et al.

Proceedings of the Conference on Empirical Methods in Natural Language Processing. Conference on Empirical Methods in Natural Language Processing · Jan 13, 2024 · 96 citations

Clinicians often rely on data engineers to retrieve complex patient information from electronic health record (EHR) systems, a process that is both inefficient and time-consuming. We propose EHRAgent, a large language model (LLM) agent empowered with accumulative domain knowledge and robust coding capability. EHRAgent enables autonomous code generation and execution to facilitate clinicians in directly interacting with EHRs using natural language. Specifically, we formulate a multi-tabular reasoning task based on EHRs as a tool-use planning process, efficiently decomposing a complex task into a sequence of manageable actions with external toolsets. We first inject relevant medical information to enable EHRAgent to effectively reason about the given query, identifying and extracting the required records from the appropriate tables. By integrating interactive coding and execution feedback, EHRAgent then effectively learns from error messages and iteratively improves its originally generated code. Experiments on three real-world EHR datasets show that EHRAgent outperforms the strongest baseline by up to 29.6% in success rate, verifying its strong capacity to tackle complex clinical tasks with minimal demonstrations.

5. · 100% match · 2025 · 4.9 cit/yr

RiskAgent: Autonomous Medical AI Copilot for Generalist Risk Prediction ([link](#))

Fenglin Liu et al.

ArXiv · Apr 11, 2025 · 6 citations

The application of Large Language Models (LLMs) to various clinical applications has attracted growing research attention. LLMs currently achieve competitive results compared to human experts in examinations. However, real-world clinical decision-making differs significantly from the standardized, exam-style scenarios commonly used in current efforts. It therefore remains a challenge to apply LLMs to complex medical tasks that require a deep understanding of medical knowledge. A common approach is to fine-tune LLMs for target tasks, which, however, not only requires substantial data and computational resources but is also still prone to generating ‘hallucinations’. In this paper, we present the RiskAgent system to perform a broad range of medical risk predictions, covering over 387 risk scenarios across diverse complex diseases, e.g., cardiovascular disease and cancer. RiskAgent is designed to collaborate with hundreds of clinical decision tools, i.e., risk calculators and scoring systems that are supported by evidence-based medicine. To evaluate our method, we have built the first benchmark MedRisk specialized for risk prediction, including 12,352 questions spanning 154 diseases, 86 symptoms, 50 specialties, and 24 organ systems. The results show that our RiskAgent, with 8 billion model parameters, achieves 76.33% accuracy, outperforming the most recent commercial LLMs, o1, o3-mini, and GPT-4.5, and doubling the 38.39% accuracy of GPT-4o. On rare diseases, e.g., Idiopathic Pulmonary Fibrosis (IPF), RiskAgent outperforms o1 and GPT-4.5 by 27.27% and 45.46% accuracy, respectively. Finally, we further conduct a generalization evaluation on an external evidence-based diagnosis benchmark and show that our RiskAgent achieves the best results. These encouraging results demonstrate the great potential of our solution for diverse diagnosis domains. For example, instead

of extensively fine-tuning LLMs for different medical tasks, our method, which collaborates with and utilizes existing evidence-based medical tools, not only achieves trustworthy results but also reduces resource costs, thus making LLMs accessible to resource-limited clinical applications. To improve the adaptability of our model in different scenarios, we have built and open-sourced a family of models ranging from 1 billion to 70 billion parameters. Our code, data, and models are all available at <https://github.com/AI-in-Health/RiskAgent>.

6. · 100% match · 2025

AKIBoards: A Structure-Following Multiagent System for Predicting Acute Kidney Injury ([link](#))

D. Gordon, P. Petousis, Susanne B. Nicholas, and Alex A. T. Bui

ArXiv · Apr 29, 2025 · 0 citations

Diagnostic reasoning entails a physician’s local (mental) model based on an assumed or known shared perspective (global model) to explain patient observations with evidence assigned towards a clinical assessment. But in several (complex) medical situations, multiple experts work together as a team to optimize health evaluation and decision-making by leveraging different perspectives. Such consensus-driven reasoning reflects individual knowledge contributing toward a broader perspective on the patient. In this light, we introduce STRUCture-following for Multiagent Systems (STRUC-MAS), a framework automating the learning of these global models and their incorporation as prior beliefs for agents in multiagent systems (MAS) to follow. We demonstrate proof of concept with a prosocial MAS application for predicting acute kidney injuries (AKIs). In this case, we found that incorporating a global structure enabled multiple agents to achieve better performance (average precision, AP) in predicting AKI 48 hours before onset (structure-following-fine-tuned, SF-FT, AP=0.195; SF-FT-retrieval-augmented generation, SF-FT-RAG, AP=0.194) vs. baseline (non-structure-following-FT, NSF-FT, AP=0.141; NSF-FT-RAG, AP=0.180) for balanced precision-weighted-recall-weighted voting. Markedly, SF-FT agents with higher recall scores reported lower confidence levels in the initial round on true positive and false negative cases. But after explicit interactions, their confidence in their decisions increased (suggesting reinforced belief). In contrast, the SF-FT agent with the lowest recall decreased its confidence in true positive and false negative cases (suggesting a new belief). This approach suggests that learning and leveraging global structures in MAS is necessary prior to achieving competitive classification and diagnostic reasoning performance.

7. · 100% match · 2025

RiskAgent: Synergizing Language Models with Validated Tools for Evidence-Based Risk Prediction ([link](#))

Fenglin Liu et al.

Mar 5, 2025 · 0 citations

Large Language Models (LLMs) achieve competitive results compared to human experts in medical examinations. However, it remains a challenge to apply LLMs to complex clinical decision-making, which requires a deep understanding of medical knowledge and differs from the standardized, exam-style scenarios commonly used in current efforts. A common approach is to fine-tune LLMs for target

tasks, which, however, not only requires substantial data and computational resources but also remains prone to generating ‘hallucinations’. In this work, we present RiskAgent, which synergizes language models with hundreds of validated clinical decision tools supported by evidence-based medicine, to provide generalizable and faithful recommendations. Our experiments show that RiskAgent not only achieves superior performance on a broad range of clinical risk predictions across diverse scenarios and diseases, but also demonstrates robust generalization in tool learning on the external MedCalc-Bench dataset, as well as in medical reasoning and question answering on three representative benchmarks, MedQA, MedMCQA, and MMLU.

8. · 100% match · 2024 · 4.1 cit/yr

Forecasting acute kidney injury and resource utilization in ICU patients using longitudinal, multimodal models ([link](#))

Yukun Tan et al.

Journal of biomedical informatics · Apr 29, 2024 · 9 citations

BACKGROUND Advances in artificial intelligence (AI) have realized the potential of revolutionizing healthcare, such as predicting disease progression via longitudinal inspection of Electronic Health Records (EHRs) and lab tests from patients admitted to Intensive Care Units (ICU). Although substantial literature exists addressing broad subjects, including the prediction of mortality, length-of-stay, and readmission, studies focusing on forecasting Acute Kidney Injury (AKI), specifically dialysis anticipation like Continuous Renal Replacement Therapy (CRRT) are scarce. The technicality of how to implement AI remains elusive.

OBJECTIVE This study aims to elucidate the important factors and methods that are required to develop effective predictive models of AKI and CRRT for patients admitted to ICU, using EHRs in the Medical Information Mart for Intensive Care (MIMIC) database.

METHODS We conducted a comprehensive comparative analysis of established predictive models, considering both time-series measurements and clinical notes from MIMIC-IV databases. Subsequently, we proposed a novel multi-modal model which integrates embeddings of top-performing unimodal models, including Long Short-Term Memory (LSTM) and BioMedBERT, and leverages both unstructured clinical notes and structured time series measurements derived from EHRs to enable the early prediction of AKI and CRRT.

RESULTS Our multimodal model achieved a lead time of at least 12 h ahead of clinical manifestation, with an Area Under the Receiver Operating Characteristic Curve (AUROC) of 0.888 for AKI and 0.997 for CRRT, as well as an Area Under the Precision Recall Curve (AUPRC) of 0.727 for AKI and 0.840 for CRRT, respectively, which significantly outperformed the baseline models. Additionally, we performed a SHapley Additive exPlanation (SHAP) analysis using the expected gradients algorithm, which highlighted important, previously underappreciated predictive features for AKI and CRRT.

CONCLUSION Our study revealed the importance and the technicality of applying longitudinal, multimodal modeling to improve early prediction of AKI and CRRT, offering insights for timely interventions. The performance and interpretability of our model indicate its potential for further

assessment towards clinical applications, to ultimately optimize AKI management and enhance patient outcomes.

9. · 100% match · 2019 · 1.5 cit/yr

Development and validation of an automated algorithm for identifying patients at higher risk for drug-induced acute kidney injury. ([link](#))

Nakyung Jeon, Benjamin J. Staley, Carl H. Henriksen, Gloria P. Lipori, and A. Winterstein

American journal of health-system pharmacy : AJHP : official journal of the American Society of Health-System Pharmacists · May 2, 2019 · 11 citations

PURPOSE Using information from institutional electronic health records, we aimed to develop dynamic predictive models to identify patients at high risk of acute kidney injury (AKI) among those who received a nephrotoxic medication during their hospital stay.

METHODS Candidate predictors were measured for each of the first 5 hospital days where a patient received a nephrotoxic medication (risk model days) to predict an AKI, using logistic regression with reduced backward variables elimination in 100 bootstrap samples. An AKI event was defined as an increase of serum creatinine $\geq 200\%$ of a baseline SCr within 5 days after a risk model day. Final models were internally validated by replication in 100 bootstrap samples and a risk score for each patient was calculated from the validated model. As performance measures, the area under the receiver operation characteristic curves (AUC) and the number of AKI events among patients who had high risk scores were estimated.

RESULTS The study population included 62,561 admissions followed by 1,212 AKI events (1.9 events/100 admissions). We constructed 5 risk models corresponding to the first 5 hospital days where patients were exposed to at least one nephrotoxic medication. Validated AUCs of the 5 models ranged from 0.78 to 0.81. Depending on risk model day, admissions ranked in the 90th percentile of the risk score captured between 43% to 49% of all AKI events.

CONCLUSION A dynamic prediction model was built successfully for inpatient AKI with excellent discriminative validity and good calibration, allowing clinicians to focus on a select high-risk population that captures the majority of AKI events.

10. · 100% match · 2024 · 21 cit/yr

ColaCare: Enhancing Electronic Health Record Modeling through Large Language Model-Driven Multi-Agent Collaboration ([link](#))

Zixiang Wang et al.

Proceedings of the ACM on Web Conference 2025 · Oct 3, 2024 · 36 citations

We introduce ColaCare, a framework that enhances Electronic Health Record (EHR) modeling through multi-agent collaboration driven by Large Language Models (LLMs). Our approach seamlessly integrates domain-specific expert models with LLMs to bridge the gap between structured EHR data and text-based reasoning. Inspired by the Multidisciplinary Team (MDT) approach used in clinical settings, ColaCare employs two types of agents: DoctorAgents and a MetaAgent, which collaboratively analyze patient data. Expert models process and generate predictions from numerical EHR data, while LLM

agents produce reasoning references and decision-making reports within the MDT-driven collaborative consultation framework. The MetaAgent orchestrates the discussion, facilitating consultations and evidence-based debates among DoctorAgents, simulating diverse expertise in clinical decision-making. We additionally incorporate the Merck Manual of Diagnosis and Therapy (MSD) medical guideline within a retrieval-augmented generation (RAG) module for medical evidence support, addressing the challenge of knowledge currency. Extensive experiments conducted on three EHR datasets demonstrate ColaCare's superior performance in clinical mortality outcome and readmission prediction tasks, underscoring its potential to revolutionize clinical decision support systems and advance personalized precision medicine. All code, case studies and a questionnaire are available at the project website: <https://colacare.netlify.app>.

11. · 100% match · 2026 · 4.0 cit/yr

Vital Trace: Protocol-Constrained Patient-State Reasoning for Longitudinal Clinical Trajectories ([link](#))

Zhan Qu and Michaela K. Farber

Feb 13, 2026 · 2 citations

Longitudinal clinical reasoning over electronic health records requires tracking evolving physiological measurements, laboratory results, and interventions across extended patient trajectories. Existing LLM-based clinical reasoning systems often rely on repeatedly serializing patient histories or exchanging unconstrained textual agent messages, leading to context drift, unstable reasoning, and growing inference cost over long horizons. We present Vital Trace, a protocol-constrained multi-agent framework for future clinical risk prediction over evolving ICU trajectories. Instead of maintaining unbounded textual histories, Vital Trace uses a compact persistent patient-state memory together with staged reasoning performed by four coordinated agents: a Router, Reasoner, Auditor, and Steward. To support temporally coherent reasoning, we introduce a manually curated Global Protocol containing physiological state-transition rules and a dynamic patient-state representation that tracks hemodynamic, respiratory, renal, metabolic, and inflammatory instability over time. We evaluate Vital Trace on MIMIC-IV and eICU using future vasopressor-support, respiratory-support, renal-support, and deterioration prediction tasks. Results show that structured protocol-constrained reasoning improves temporal consistency, communication stability, calibration, and interpretability compared with free-form multi-agent baselines while achieving strong predictive performance across long ICU trajectories.

12. · 100% match · 2024 · 3.4 cit/yr

Predicting Nephrotoxic Acute Kidney Injury in Hospitalized Adults: A Machine Learning Algorithm ([link](#))

Benjamin R. Griffin et al.

Kidney Medicine · Oct 1, 2024 · 6 citations

Rationale and Objective Acute kidney injury (AKI) is a common complication among hospitalized adults, but AKI prediction and prevention among adults has proved challenging. We used machine learning to update the nephrotoxic injury negated by just-in time action (NINJA), a pediatric program

that predicts nephrotoxic AKI, to improve accuracy among adults. **Study Design** A retrospective cohort study. **Setting and Population** Adults admitted for > 48 hours to the University of Iowa Hospital from 2017 to 2022. **Exposure** A NINJA high-nephrotoxin exposure (≥ 3 nephrotoxins on 1 day or intravenous aminoglycoside or vancomycin for ≥ 3 days). **Outcomes** AKI within 48 hours of high-nephrotoxin exposure. **Analytical Approach** We collected 85 variables, including demographics, laboratory tests, vital signs, and medications. AKI was defined as a serum creatinine increase of ≥ 0.3 mg/dL. A gated recurrent unit (GRU)-based recurrent neural network (RNN) was trained on 85% of the data, and then tested on the remaining 15%. Model performance was evaluated with precision, recall, negative predictive value, and area under the curve. We used an artificial neural network to determine risk factor importance. **Results** There were 14,480 patients, 18,180 admissions, and 37,300 high-nephrotoxin exposure events meeting inclusion criteria. In the testing cohort, 29% of exposures developed AKI within 48 hours. The RNN-GRU model predicted AKI with a precision of 0.60, reducing the number of false alerts from 2.5 to 0.7 per AKI case. Lowest hemoglobin, lowest blood pressure, and highest white blood cell count were the most important variables in the artificial neural network model. Acyclovir, piperacillin-tazobactam, calcineurin inhibitors, and angiotensin-converting enzyme inhibitor/angiotensin receptor blockers were the most important medications. **Limitations** Clinical variables and medications were not exhaustive, drug levels or dosing were not incorporated, and Iowa's racial makeup may limit generalizability. **Conclusions** Our RNN-GRU model substantially reduced the number of false alerts for nephrotoxic AKI, which may facilitate NINJA translation to adult hospitals by providing more targeted intervention.

13. · 100% match · 2023 · 16 cit/yr

Reliable generation of privacy-preserving synthetic electronic health record time series via diffusion models ([link](#))

Muhang Tian, Bernie Chen, Allan Guo, Shiyi Jiang, and Anru R. Zhang

Journal of the American Medical Informatics Association : JAMIA · Oct 23, 2023 · 44 citations

OBJECTIVE Electronic health records (EHRs) are rich sources of patient-level data, offering valuable resources for medical data analysis. However, privacy concerns often restrict access to EHRs, hindering downstream analysis. Current EHR deidentification methods are flawed and can lead to potential privacy leakage. Additionally, existing publicly available EHR databases are limited, preventing the advancement of medical research using EHR. This study aims to overcome these challenges by generating realistic and privacy-preserving synthetic EHRs time series efficiently.

MATERIALS AND METHODS We introduce a new method for generating diverse and realistic synthetic EHR time series data using denoizing diffusion probabilistic models. We conducted experiments on 6 databases: Medical Information Mart for Intensive Care III and IV, the eICU Collaborative Research Database (eICU), and non-EHR datasets on Stocks and Energy. We compared our proposed method with 8 existing methods.

RESULTS Our results demonstrate that our approach significantly outperforms all existing methods in terms of data fidelity while requiring less training effort. Additionally, data generated by our method yield a lower discriminative accuracy compared to other baseline methods, indicating the proposed method can generate data with less privacy risk.

DISCUSSION The proposed model utilizes a mixed diffusion process to generate realistic synthetic EHR samples that protect patient privacy. This method could be useful in tackling data availability issues in the field of healthcare by reducing barrier to EHR access and supporting research in machine learning for health.

CONCLUSION The proposed diffusion model-based method can reliably and efficiently generate synthetic EHR time series, which facilitates the downstream medical data analysis. Our numerical results show the superiority of the proposed method over all other existing methods.

14. · 100% match · 2025 · 4.4 cit/yr

MoMA: a mixture-of-multimodal-agents architecture for enhancing clinical prediction modeling ([link](#))

Jifan Gao et al.

NPJ Digital Medicine · Aug 7, 2025 · 4 citations

Multimodal electronic health record (EHR) data provide richer, complementary insights into patient health compared to single-modality data. However, effectively integrating diverse data modalities for clinical prediction modeling remains challenging due to the substantial data requirements. We introduce a novel architecture, Mixture-of-Multimodal-Agents (MoMA), designed to leverage multiple large language model (LLM) agents for clinical prediction tasks using multimodal EHR data. MoMA employs specialized LLM agents (“specialist agents”) to convert non-textual modalities, such as medical images and laboratory results, into structured textual summaries. These summaries, together with clinical notes, are combined by another LLM (“aggregator agent”) to generate a unified multimodal summary, which is then used by a third LLM (“predictor agent”) to produce clinical predictions. Evaluating MoMA with different modality combinations and prediction settings, MoMA outperforms existing methods on three prediction tasks using private datasets, highlighting its enhanced accuracy and flexibility across various tasks.

15. · 100% match · 2026

TreeText-CTS: Compact, Source-Traceable Tree-Path Evidence for Irregular Clinical Time-Series Prediction ([link](#))

Kwanhyung Lee et al.

May 19, 2026 · 0 citations

Numerical time-series models can effectively process irregular electronic health record (EHR) trajectories, but they do not naturally expose the measurements and temporal patterns supporting each risk estimate as readable evidence. Existing text-based interfaces improve readability, but typically rely on either raw serialization, which is lengthy and redundant, or patient-level free-form summaries, which are difficult to trace to source measurements and time windows. To bridge this gap, we introduce TreeText-CTS (Clinical Time-Series), which converts irregular EHR trajectories into human-readable, compact, source-traceable tree-path evidence units without patient-level summarization or inference-time autoregressive decoding. TreeText-CTS routes multi-scale window summaries through frozen XGBoost models and verbalizes activated tree paths as deterministic, source-traceable evidence units

composed of threshold conditions. An evidence selector assembles an informative subset of these units, which a language-model encoder then integrates for prediction. Across PhysioNet 2012 mortality, MIMIC-III mortality, and PhysioNet 2019 sepsis-onset forecasting, TreeText-CTS achieves the best AUROC and AUPRC among evaluated text-based EHR time-series interfaces, improving AUPRC by 6.0 to 9.7 absolute percentage points over the strongest prior text-based interface while remaining competitive with numerical time-series models. Ablations show that tree-path evidence construction, evidence selection, and language-model composition each contribute to performance. Because every span passed to the language-model encoder is constructed from activated tree-path threshold conditions, TreeText-CTS makes the evidence supplied to the final predictor inspectable and source-traceable.

16. · 100% match · 2026

Beyond binary AKI classification: development and external validation of a distributional model predicting serum creatinine and urine output trajectories in ICU patients ([link](#))

Jef Jonkers et al.

Critical Care · May 10, 2026 · 0 citations

Acute kidney injury (AKI) is a frequent, severe complication in the intensive care units (ICU). Existing machine learning models are typically inflexible, classification-based (i.e., predicting AKI occurrence as yes/no), and of limited clinical utility. This study proposes and externally validates the first multi-step, multivariate distributional regression model that directly predicts future distributions of serum creatinine (sCr) and urine output across multiple time horizons, thereby enhancing AKI risk stratification and personalized clinical decision support. The model was developed using a training cohort of 4,118 adult ICU stays from the MIMIC-IV dataset and externally validated on four independent, diverse cohorts: MIMIC-IV (N=3,838), UZGent (N=4,442), eICU (N=10,760), and AmsterdamUMC (N=6,129). The model used clinical data to generate multivariate predictive distributions hourly for urine output and sCr (up to 48 hours ahead). Predictors included demographics, vital signs, laboratory results, medications, and recent urine output, with time-varying variables summarized over the preceding 72 hours (recent value, slope, minimum, maximum, variability). Performance was evaluated by comparing our predictive distributions with state-of-the-art tree-based classifiers for 24-hour ahead prediction of KDIGO stages 1-3 AKI and persistent stage 3 AKI. Across all external cohorts, the distributional regression model demonstrated high discrimination (mean AUC-PR 0.774 for all stages) and excellent calibration, consistently outperforming the benchmark classifiers. By jointly predicting sCr and urine output distributions, a single model successfully enables flexible risk stratification across all stages, capturing AKI onset and persistence, and allowing changes to stage definitions. This multi-step, multivariate distributional regression model is a reliable, more flexible, transparent, and clinically interpretable approach for AKI prediction compared to traditional classification methods. It represents a necessary step toward bedside implementation of predictive models for personalized AKI management in the ICU.

17. · 100% match · 2025 · 2.5 cit/yr

EHR-MCP: Real-world Evaluation of Clinical Information Retrieval by Large Language Models via Model Context Protocol ([link](#))

Kanato Masayoshi et al.

ArXiv · Sep 19, 2025 · 2 citations

Background: Large language models (LLMs) show promise in medicine, but their deployment in hospitals is limited by restricted access to electronic health record (EHR) systems. The Model Context Protocol (MCP) enables integration between LLMs and external tools. Objective: To evaluate whether an LLM connected to an EHR database via MCP can autonomously retrieve clinically relevant information in a real hospital setting. Methods: We developed EHR-MCP, a framework of custom MCP tools integrated with the hospital EHR database, and used GPT-4.1 through a LangGraph ReAct agent to interact with it. Six tasks were tested, derived from use cases of the infection control team (ICT). Eight patients discussed at ICT conferences were retrospectively analyzed. Agreement with physician-generated gold standards was measured. Results: The LLM consistently selected and executed the correct MCP tools. Except for two tasks, all tasks achieved near-perfect accuracy. Performance was lower in the complex task requiring time-dependent calculations. Most errors arose from incorrect arguments or misinterpretation of tool results. Responses from EHR-MCP were reliable, though long and repetitive data risked exceeding the context window. Conclusions: LLMs can retrieve clinical data from an EHR via MCP tools in a real hospital setting, achieving near-perfect performance in simple tasks while highlighting challenges in complex ones. EHR-MCP provides an infrastructure for secure, consistent data access and may serve as a foundation for hospital AI agents. Future work should extend beyond retrieval to reasoning, generation, and clinical impact assessment, paving the way for effective integration of generative AI into clinical practice.

18. · 100% match · 2023 · 1251 cit/yr

Toolformer: Language Models Can Teach Themselves to Use Tools ([link](#))

Timo Schick et al.

ArXiv · Feb 9, 2023 · 4249 citations

Language models (LMs) exhibit remarkable abilities to solve new tasks from just a few examples or textual instructions, especially at scale. They also, paradoxically, struggle with basic functionality, such as arithmetic or factual lookup, where much simpler and smaller models excel. In this paper, we show that LMs can teach themselves to use external tools via simple APIs and achieve the best of both worlds. We introduce Toolformer, a model trained to decide which APIs to call, when to call them, what arguments to pass, and how to best incorporate the results into future token prediction. This is done in a self-supervised way, requiring nothing more than a handful of demonstrations for each API. We incorporate a range of tools, including a calculator, a Q&A system, two different search engines, a translation system, and a calendar. Toolformer achieves substantially improved zero-shot performance across a variety of downstream tasks, often competitive with much larger models, without sacrificing its core language modeling abilities.

19. · 100% match · 2026

Forecasting ICU Acute Kidney Injury with Actionable Lead Time Using Interpretable Machine Learning: Development and Multi-Center Validation ([link](#))

A. Hourani, Z. Jakubowska, and J. Małyszko

Journal of Clinical Medicine · Feb 1, 2026 · 0 citations

Background: Acute kidney injury (AKI) is common in intensive care and is frequently recognized late. We aimed to develop an interpretable, dynamic early warning score (EWS) to predict incident AKI within the next 24 h in ICU adult patients and to test its transportability temporally and geographically. Methods: We performed a retrospective cohort multi-center study using hospitalized ICU patients from MIMIC-IV and eICU-CRD databases. The outcome was incident AKI (KDIGO stage 1–3). Gradient-boosted trees (XGBoost) were trained with 10-fold cross-validation. Predictors were prespecified from the literature and finalized as 61 routinely available EHR predictors selected using SHAP ranking (spanning demographics/comorbidities, laboratory markers, vital-sign dynamics, and ICU therapies/procedures). Prespecified validations included (i) temporal validation in the COVID-19 era and (ii) geographic validation in eICU Northeast hospitals. Results: The development cohort included 51,833 ICU stays; temporal and geographic cohorts included 3346 and 2993 stays, respectively. Discrimination was high in internal validation (AUC 0.88, 95% CI 0.84–0.90) and remained robust temporally (AUC 0.84, 0.80–0.87) and geographically (AUC 0.82, 0.81–0.84). At a prespecified threshold, sensitivity/specificity were 0.76/0.79 (temporal) and 0.73/0.86 (geographic). Decision-curve analysis showed net benefit across clinically relevant thresholds. Key predictors reflected physiologic trajectories (e.g., mean arterial pressure dynamics), urine output, renal/metabolic markers (e.g., creatinine and BUN trends), and oxygenation dynamics (SpO₂). Conclusions: A routinely updated, explainable EHR-based EWS can predict ICU AKI up to 24 h in advance with stable performance across a pandemic-era temporal shift and external geographic validation.

20. · 100% match · 2024 · 0.8 cit/yr

Enhancing Acute Kidney Injury Prediction through Integration of Drug Features in Intensive Care Units ([link](#))

Gabriel D. M. Manalu, Mulomba Mukendi Christian, Songhee You, and Hyebyong Choi

ArXiv · Jan 9, 2024 · 2 citations

The relationship between acute kidney injury (AKI) prediction and nephrotoxic drugs, or drugs that adversely affect kidney function, is one that has yet to be explored in the critical care setting. One contributing factor to this gap in research is the limited investigation of drug modalities in the intensive care unit (ICU) context, due to the challenges of processing prescription data into the corresponding drug representations and a lack in the comprehensive understanding of these drug representations. This study addresses this gap by proposing a novel approach that leverages patient prescription data as a modality to improve existing models for AKI prediction. We base our research on Electronic Health Record (EHR) data, extracting the relevant patient prescription information and converting it into the selected drug representation for our research, the extended-connectivity fingerprint (ECFP). Furthermore, we adopt a unique multimodal approach, developing machine learning models and 1D Convolutional Neural Networks (CNN) applied to clinical drug representations, establishing a procedure which has not been used by any previous studies predicting AKI. The findings showcase a notable improvement in AKI prediction through the integration of drug embeddings and other patient cohort features. By using drug features represented as ECFP molecular fingerprints along with common cohort

features such as demographics and lab test values, we achieved a considerable improvement in model performance for the AKI prediction task over the baseline model which does not include the drug representations as features, indicating that our distinct approach enhances existing baseline techniques and highlights the relevance of drug data in predicting AKI in the ICU setting

21. · 100% match · 2023 · 2.1 cit/yr

Time Series AI Model for Acute Kidney Injury Detection Based on a Multicenter Distributed Research Network: Development and Verification Study ([link](#))

S. Heo et al.

JMIR Medical Informatics · Mar 30, 2023 · 7 citations

Abstract Background Acute kidney injury (AKI) is a marker of clinical deterioration and renal toxicity. While there are many studies offering prediction models for the early detection of AKI, those predicting AKI occurrence using distributed research network (DRN)-based time series data are rare. **Objective** In this study, we aimed to detect the early occurrence of AKI by applying an interpretable long short-term memory (LSTM)-based model to hospital electronic health record (EHR)-based time series data in patients who took nephrotoxic drugs using a DRN. **Methods** We conducted a multi-institutional retrospective cohort study of data from 6 hospitals using a DRN. For each institution, a patient-based data set was constructed using 5 drugs for AKI, and an interpretable multivariable LSTM (IMV-LSTM) model was used for training. This study used propensity score matching to mitigate differences in demographics and clinical characteristics. Additionally, the temporal attention values of the AKI prediction model's contribution variables were demonstrated for each institution and drug, with differences in highly important feature distributions between the case and control data confirmed using 1-way ANOVA. **Results** This study analyzed 8643 and 31,012 patients with and without AKI, respectively, across 6 hospitals. When analyzing the distribution of AKI onset, vancomycin showed an earlier onset (median 12, IQR 5-25 days), and acyclovir was the slowest compared to the other drugs (median 23, IQR 10-41 days). Our temporal deep learning model for AKI prediction performed well for most drugs. Acyclovir had the highest average area under the receiver operating characteristic curve score per drug (0.94), followed by acetaminophen (0.93), vancomycin (0.92), naproxen (0.90), and celecoxib (0.89). Based on the temporal attention values of the variables in the AKI prediction model, verified lymphocytes and calcvancomycin ium had the highest attention, whereas lymphocytes, albumin, and hemoglobin tended to decrease over time, and urine pH and prothrombin time tended to increase. **Conclusions** Early surveillance of AKI outbreaks can be achieved by applying an IMV-LSTM based on time series data through an EHR-based DRN. This approach can help identify risk factors and enable early detection of adverse drug reactions when prescribing drugs that cause renal toxicity before AKI occurs.

22. · 100% match · 2021 · 28 cit/yr

Generating synthetic mixed-type longitudinal electronic health records for artificial intelligent applications ([link](#))

Jin Li, B. Cairns, Jingsong Li, and T. Zhu

NPJ Digital Medicine · Dec 22, 2021 · 127 citations

The recent availability of electronic health records (EHRs) have provided enormous opportunities to develop artificial intelligence (AI) algorithms. However, patient privacy has become a major concern that limits data sharing across hospital settings and subsequently hinders the advances in AI. Synthetic data, which benefits from the development and proliferation of generative models, has served as a promising substitute for real patient EHR data. However, the current generative models are limited as they only generate single type of clinical data for a synthetic patient, i.e., either continuous-valued or discrete-valued. To mimic the nature of clinical decision-making which encompasses various data types/sources, in this study, we propose a generative adversarial network (GAN) entitled EHR-M-GAN that simultaneously synthesizes mixed-type timeseries EHR data. EHR-M-GAN is capable of capturing the multidimensional, heterogeneous, and correlated temporal dynamics in patient trajectories. We have validated EHR-M-GAN on three publicly-available intensive care unit databases with records from a total of 141,488 unique patients, and performed privacy risk evaluation of the proposed model. EHR-M-GAN has demonstrated its superiority over state-of-the-art benchmarks for synthesizing clinical timeseries with high fidelity, while addressing the limitations regarding data types and dimensionality in the current generative models. Notably, prediction models for outcomes of intensive care performed significantly better when training data was augmented with the addition of EHR-M-GAN-generated timeseries. EHR-M-GAN may have use in developing AI algorithms in resource-limited settings, lowering the barrier for data acquisition while preserving patient privacy.

23. · 100% match · 2023 · 30 cit/yr

EHR-Safe: generating high-fidelity and privacy-preserving synthetic electronic health records
([link](#))

Jinsung Yoon et al.

NPJ Digital Medicine · Aug 11, 2023 · 88 citations

Privacy concerns often arise as the key bottleneck for the sharing of data between consumers and data holders, particularly for sensitive data such as Electronic Health Records (EHR). This impedes the application of data analytics and ML-based innovations with tremendous potential. One promising approach for such privacy concerns is to instead use synthetic data. We propose a generative modeling framework, EHR-Safe, for generating highly realistic and privacy-preserving synthetic EHR data. EHR-Safe is based on a two-stage model that consists of sequential encoder-decoder networks and generative adversarial networks. Our innovations focus on the key challenging aspects of real-world EHR data: heterogeneity, sparsity, coexistence of numerical and categorical features with distinct characteristics, and time-varying features with highly-varying sequence lengths. Under numerous evaluations, we demonstrate that the fidelity of EHR-Safe is almost-identical with real data (<3% accuracy difference for the models trained on them) while yielding almost-ideal performance in practical privacy metrics.

24. · 100% match · 2025 · 15 cit/yr

Artificial intelligence models for predicting acute kidney injury in the intensive care unit: a systematic review of modeling methods, data utilization, and clinical applicability ([link](#))

Tongyue Shi, Yu Lin, Huiying Zhao, and Guilan Kong

JAMIA Open · Jul 3, 2025 · 15 citations

Abstract Objectives Acute kidney injury (AKI) is common in intensive care unit (ICU) patients and is associated with high mortality, prolonged ICU stays, and increased costs. Early prediction is crucial for timely intervention and improved outcomes. Various prediction models, including machine learning, deep learning, and dynamic prediction frameworks, have been developed, but their modeling approaches, data utilization, and clinical applicability require further investigation. This review comprehensively assesses the modeling methods, data utilization strategies, and clinical applicability of AKI prediction models in the ICU, identifies current challenges, and proposes future research directions. **Materials and Methods** A systematic search was conducted in PubMed, Embase, Scopus, Web of Science, IEEE Xplore, and ACM Digital Library up to December 12, 2024. Studies were included if they reported AKI prediction models using ICU-specific data, included at least 2 predictors, and evaluated model performance. Extracted data included study characteristics, model details, data sources, performance metrics, and validation methods. The risk of bias was assessed using PROBAST (Prediction Model Risk of Bias Assessment Tool), and the reporting quality was evaluated using the TRIPOD (Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis) guideline. **Results** From 1305 screened studies, 47 met the inclusion criteria. Models ranged from machine learning to advanced deep learning techniques. Only 14 studies conducted external validation. Most studies (n=44) had a high risk of bias, particularly in generalizability and clinical applicability. **Discussion** Although AI models have shown promise in predicting AKI in ICU settings, key challenges remain. These include limited external validation, lack of dynamic modeling, insufficient interpretability, and poor consideration of clinical integration. Different study designs, prediction windows, and data sources also hinder model comparability. **Conclusions** Future research should prioritize dynamic, interpretable, and externally validated models. These efforts are critical to bridge the gap between model development and clinical implementation and to enhance the real-world applicability of AI in AKI prediction.

25. · 100% match · 7 cit/yr

EHRFlow : A Large Language Model-Driven Iterative Multi-Agent Electronic Health Record Data Analysis Workflow ([link](#))

Hao Wu et al.

7 citations

EHRFlow is a large language model-driven platform that simplifies electronic health record (EHR) data analysis for physicians through natural language interactions, eliminating the need for complex coding. EHRFlow integrates four key agents: PlanAgent, ToolAgent, CodeAgent, and ReviewAgent in a dual-loop mechanism that efficiently decomposes and executes tasks via PlanAgent, from coarse-grained conceptual planning to fine-grained task execution, leveraging a healthcare tool bank and the code generation and execution capabilities of ToolAgent and CodeAgent. The ReviewAgent ensures the safety and quality of code through iterative feedback and refinement. EHRFlow offers a user-friendly solution for physicians' customized needs while preserving sensitive data's privacy by executing code

operations locally. Demonstrated through case studies, EHRFlow showcases its ability to coordinate agents in resolving complex medical data analysis tasks, highlighting its flexibility,

26. · 100% match · 2019 · 133 cit/yr

A Clinically Applicable Approach to Continuous Prediction of Future Acute Kidney Injury ([link](#))

Nenad Tomašev et al.

Nature · Jul 31, 2019 · 921 citations

The early prediction of deterioration could have an important role in supporting healthcare professionals, as an estimated 11% of deaths in hospital follow a failure to promptly recognize and treat deteriorating patients¹. To achieve this goal requires predictions of patient risk that are continuously updated and accurate, and delivered at an individual level with sufficient context and enough time to act. Here we develop a deep learning approach for the continuous risk prediction of future deterioration in patients, building on recent work that models adverse events from electronic health records^{2–17} and using acute kidney injury—a common and potentially life-threatening condition¹⁸—as an exemplar. Our model was developed on a large, longitudinal dataset of electronic health records that cover diverse clinical environments, comprising 703,782 adult patients across 172 inpatient and 1,062 outpatient sites. Our model predicts 55.8% of all inpatient episodes of acute kidney injury, and 90.2% of all acute kidney injuries that required subsequent administration of dialysis, with a lead time of up to 48 h and a ratio of 2 false alerts for every true alert. In addition to predicting future acute kidney injury, our model provides confidence assessments and a list of the clinical features that are most salient to each prediction, alongside predicted future trajectories for clinically relevant blood tests⁹. Although the recognition and prompt treatment of acute kidney injury is known to be challenging, our approach may offer opportunities for identifying patients at risk within a time window that enables early treatment. A deep learning approach that predicts the risk of acute kidney injury may help to identify patients at risk of health deterioration within a time window that enables early treatment.

27. · 100% match · 2026

AKI-twinX: explainable organ structured digital twin for sepsis AKI trajectory forecasting ([link](#))

Jinjin Cai et al.

medRxiv · Mar 22, 2026 · 0 citations

Acute kidney injury in sepsis evolves over hours to days, yet most ICU models emphasize onset and provide limited insight into cardio-renal interactions. We developed AKI-twinX, an organ-structured, explainable digital twin that jointly forecasts acute kidney injury onset, acute kidney injury trajectory, and near-term mortality risk. The model learns renal and cardiovascular latent states with sparse feature gating and captures cross-organ coupling with attention. We trained AKI-twinX on MIMIC-IV sepsis using 5-fold cross-validation and evaluated it on an Indiana University Health cohort. Discrimination was consistent across systems (AUC: mortality 0.86-0.88, acute kidney injury onset 0.78-0.82, acute kidney injury trajectory 0.73-0.78). In vasopressor-treated windows, 12-hour systolic blood pressure forecasts tracked observed values (mean absolute error 8.5 mmHg). Counterfactual vasopressor withdrawal shifted predicted blood pressure downward and increased predicted risk,

supporting sensitivity to clinically meaningful interventions. AKI-twinX enables trajectory-aware forecasting with bedside auditability in sepsis.

28. · 100% match · 2026

Machine learning studies of drug-induced nephrotoxicity: a scoping review ([link](#))

Mawardi Ihsan, Shun-Ting Chang, Wei-Kai Chan, and Hsiang-Yin Chen

Therapeutic Advances in Drug Safety · Mar 1, 2026 · 0 citations

Background: Machine learning methods have emerged as a promising approach to prevent drug-induced nephrotoxicity. Objective: This review evaluates the quality and highlights recent advances of machine learning algorithms for predicting drug-induced nephrotoxicity. Eligibility criteria: Studies on machine learning models to predict drug-induced acute kidney injury, acute kidney disease, or both published between January 2014 and August 2024 were eligible. Sources of evidence: A comprehensive search was conducted by using PubMed, Embase, Web of Science, Cochrane Library, and Scopus. Charting methods: A standardized charting form was developed based on CHARMs, TRIPOD+AI, and PROBAST tools to assess the quality and risk of bias across studies. Results: From the initial 5,179 articles searched, 24 studies were included in this review. All studies achieved good area under the receiver operating characteristic curves (AUROCs) above 0.75, with boosting machines being the most frequently outperforming algorithms ($n = 7$, 29.17%), and neural networks showed the highest median AUROC of 0.90 (0.86–0.92). Two-thirds of studies ($n = 16$; 66.67%) predicted acute kidney injury, whereas only 5 (20.83%) focused on acute kidney disease. Estimated glomerular filtration rate, blood urea nitrogen, serum creatinine, hemoglobin, and albumin emerged as the most utilized features by 10 (41.67%), 9 (37.5%), 9 (37.5%), 8 (33.33%), and 8 (33.33%) studies, respectively. Diabetes, heart failure, diuretics, and non-steroidal anti-inflammatory drugs were frequently selected features by 7 (29.17%), 5 (20.83%), 5 (20.83%), and 4 (16.67%) studies, respectively. The 2025 PROBAST+AI risk-of-bias assessment indicated that 7 (29.17%) studies had a low risk of bias. A high risk of bias was observed in 20 (83.33%), 18 (75%), and 17 (70.83%) studies due to insufficient performance evaluation, small sample sizes, and lack of external validation. Conclusion: Recent machine learning studies have demonstrated great performance using clinically obtainable features. Incorporating acute kidney injury and disease, methodological enhancement, and guideline adherence can facilitate clinical applicability in preventing drug-induced nephrotoxicity.

29. · 100% match · 2025

Predicting Acute Kidney Injury with Nephrotoxic Burden in Noncritical Patients ([link](#))

Esra Adiyekke et al.

Kidney360 · Nov 12, 2025 · 0 citations

Visual Abstract

30. · 100% match · 2026

D2MDT: Department-aware Multidisciplinary Team Consultation with Deliberation for Efficient Clinical Prediction ([link](#))

Yongqi Liang et al.

Jun 2, 2026 · 0 citations

Electronic health records (EHRs) are central to clinical prediction, but existing methods either rely on correlation-driven deep models or use single large language models (LLMs), making it difficult to support multidisciplinary clinical reasoning. Recent multi-agent systems (MAS) provide a promising alternative, yet current EHR-grounded MAS methods still suffer from weak evidence differentiation across agents and redundant multi-round interaction. We propose D2MDT, a Department-aware MultiDisciplinary Team Consultation with Deliberation for Efficient clinical prediction. D2MDT first constructs structured EHR evidence and consultation-ready semantic evidence for multi-agent consultation. It then assigns patient-specific department perspectives to doctor agents and retrieves complementary evidence for collaborative consultation. To improve efficiency, D2MDT further introduces residual deliberation, which updates only unresolved consensus rather than replaying the full discussion history. Finally, D2MDT fuses the refined consensus report with structured EHR representations for prediction. Experiments on mortality prediction show that D2MDT improves both predictive performance and consultation efficiency. We release the code online to ease the reproducibility of this paper.

31. · 100% match · 2026

Agentifying Patient Dynamics within LLMs through Interacting with Clinical World Model
([link](#))

Minghao Wu et al.

May 14, 2026 · 0 citations

Sepsis management in the ICU requires sequential treatment decisions under rapidly evolving patient physiology. Although large language models (LLMs) encode broad clinical knowledge and can reason over guidelines, they are not inherently grounded in action-conditioned patient dynamics. We introduce SepsisAgent, a world model-augmented LLM agent for sepsis treatment recommendation. SepsisAgent uses a learned Clinical World Model to simulate patient responses under candidate fluid-vasopressor interventions, and follows a propose-simulate-refine workflow before committing to a prescription. We first show that world-model access alone yields inconsistent LLM decision performance, motivating agent-specific training. We then train SepsisAgent through a three-stage curriculum: patient-dynamics supervised fine-tuning, propose-simulate-refine behavior cloning, and world-model-based agentic reinforcement learning. On MIMIC-IV sepsis trajectories, SepsisAgent outperforms all traditional RL and LLM-based baselines in off-policy value while achieving the best safety profile under guideline adherence and unsafe-action metrics. Further analysis shows that repeated interaction with the Clinical World Model enables the agent to learn regularities in patient evolution, which remain useful even when simulator access is removed.

32. · 100% match · 2026

Computer-Interpretable Domain Knowledge for Drug-Induced Acute Kidney Injury: a Knowledge Graph Approach. ([link](#))

Romy Vos et al.

Scientific data · Jun 10, 2026 · 0 citations

33. · 100% match · 2022 · 50 cit/yr

TALM: Tool Augmented Language Models ([link](#))

Aaron Parisi, Yao Zhao, and Noah Fiedel

ArXiv · May 24, 2022 · 204 citations

Transformer based language models (LMs) demonstrate increasing performance with scale across a wide variety of tasks. Scale alone however cannot enable models to solve tasks that require access to ephemeral, changing, or private data that was unavailable at training time. Many useful tasks may also benefit from LMs being able to access APIs that read or modify state. In this work, we present Tool Augmented Language Models (TALM), combining a text-only approach to augment language models with non-differentiable tools, and an iterative "self-play" technique to bootstrap performance starting from few tool demonstrations. TALM exhibits strong performance on both a knowledge-heavy QA task and a reasoning oriented math task with simple tools. At a given model scale, TALM significantly outperforms non-augmented LMs. We further demonstrate that TALM successfully performs out-of-distribution inferences on both QA and math tasks, where non-augmented LMs fail. Our results suggest that Tool Augmented Language Models are a promising direction to enrich LMs' capabilities, with less dependence on scale.

34. · 100% match · 2025 · 29 cit/yr

Large language model agents can use tools to perform clinical calculations ([link](#))

Alex J. Goodell, Simon N. Chu, D. Rouholiman, and Larry F. Chu

NPJ Digital Medicine · Mar 17, 2025 · 38 citations

Large language models (LLMs) can answer expert-level questions in medicine but are prone to hallucinations and arithmetic errors. Early evidence suggests LLMs cannot reliably perform clinical calculations, limiting their potential integration into clinical workflows. We evaluated ChatGPT's performance across 48 medical calculation tasks, finding incorrect responses in one-third of trials (n=212). We then assessed three forms of agentic augmentation: retrieval-augmented generation, a code interpreter tool, and a set of task-specific calculation tools (OpenMedCalc) across 10,000 trials. Models with access to task-specific tools showed the greatest improvement, with LLaMa and GPT-based models demonstrating a 5.5-fold (88% vs 16%) and 13-fold (64% vs 4.8%) reduction in incorrect responses, respectively, compared to the unimproved models. Our findings suggest that integration of machine-readable, task-specific tools may help overcome LLMs' limitations in medical calculations.

35. · 100% match · 2024 · 5.4 cit/yr

An empirical study on KDIGO-defined acute kidney injury prediction in the intensive care unit ([link](#))

Xinrui Lyu et al.

Bioinformatics · Feb 3, 2024 · 13 citations

Motivation: Acute kidney injury (AKI) is a syndrome that affects a large fraction of all critically ill patients, and early diagnosis to receive adequate treatment is as imperative as it is challenging to make early. Consequently, machine learning approaches have been developed to predict AKI ahead of time. However, the prevalence of AKI is often underestimated in state-of-the-art approaches, as they rely on an AKI event annotation solely based on creatinine, ignoring urine output. Methods: We construct and evaluate early warning systems for AKI in a multi-disciplinary ICU setting, using the complete KDIGO definition of AKI. We propose several variants of gradient-boosted decision trees (GBDT)-based models, including a novel time-stacking based approach. A state-of-the-art LSTM-based model previously proposed for AKI prediction is used as a comparison, which was not specifically evaluated in ICU settings yet. Results: We find that optimal performance is achieved by using GBDT with the time-based stacking technique (AUPRC=65.7%, compared with the LSTM-based model's AUPRC=62.6%), which is motivated by the high relevance of time since ICU admission for this task. Both models show mildly reduced performance in the limited training data setting, perform fairly across different subcohorts, and exhibit no issues in gender transfer. Conclusion: Following the official KDIGO definition substantially increases the number of annotated AKI events. In our study GBDTs outperform LSTM models for AKI prediction. Generally, we find that both model types are robust in a variety of challenging settings arising for ICU data.

36. · 100% match · 2024 · 4.9 cit/yr

Machine Learning Algorithms to Predict Colistin-Induced Nephrotoxicity from Electronic Health Records in Patients with Multidrug-Resistant Gram-Negative Infection. ([link](#))

Ling-Wan Chiu et al.

International journal of antimicrobial agents · Apr 1, 2024 · 11 citations

OBJECTIVES Colistin-induced nephrotoxicity prolongs hospitalization and increases mortality. The study aimed to construct machine learning models to predict colistin-induced nephrotoxicity in patients with multidrug-resistant gram-negative infection.

METHODS Patients receiving colistin from three hospitals in the Clinical Research Database were included. Data were divided into a derivation cohort (2011~2017) and a temporal validation cohort (2018~2020). Fifteen machine learning models were established by categorical boosting, light gradient boosting machine, and random forest. Classifier performances were compared by the sensitivity, F1 score, Matthews correlation coefficient (MCC), area under the receiver operating characteristic (AUROC) curve, and area under the precision-recall curve (AUPRC). SHapley Additive exPlanations plots were drawn to understand feature importance and interactions.

RESULTS The study included 1392 patients, with 360 (36.4%) and 165 (40.9%) experiencing nephrotoxicity in the derivation and temporal validation cohorts, respectively. The categorical boosting with oversampling achieved the highest performance with a sensitivity of 0.860, an F1 score of 0.740, an MCC of 0.533, an AUROC curve of 0.823, and an AUPRC of 0.737. The feature importance demonstrated that the days of colistin use, cumulative dose, daily dose, latest C-reactive protein, and baseline hemoglobin were the most important risk factors, especially for vulnerable patients. A cutoff colistin dose of 4.0 mg/kg body weight/day was identified for patients at higher risk of nephrotoxicity.

CONCLUSIONS Machine learning techniques can be an early identification tool to predict colistin-induced nephrotoxicity. The observed interactions suggest a modification in dose adjustment guidelines. Future geographic and prospective validation studies are warranted to strengthen the real-world applicability.

37. · 100% match · 2024 · 3.5 cit/yr

IGAMT: Privacy-Preserving Electronic Health Record Synthesization with Heterogeneity and Irregularity ([link](#))

Wenjie Wang et al.

Proceedings of the ... AAAI Conference on Artificial Intelligence. AAAI Conference on Artificial Intelligence · Mar 24, 2024 · 8 citations

Utilizing electronic health records (EHR) for machine learning-driven clinical research has great potential to enhance outcome predictions and treatment personalization. Nonetheless, due to privacy and security concerns, the secondary use of EHR data is regulated, constraining researchers' access to EHR data. Generating synthetic EHR data with deep learning methods is a viable and promising approach to mitigate privacy concerns, offering not only a supplementary resource for downstream applications but also sidestepping the privacy risks associated with real patient data. While prior efforts have concentrated on EHR data synthesis, significant challenges persist: addressing the heterogeneity of features including temporal and non-temporal features, structurally missing values, and irregularity of the temporal measures, and ensuring rigorous privacy of the real data used for model training. Existing works in this domain only focused on solving one or two aforementioned challenges. In this work, we propose IGAMT, an innovative framework to generate privacy-preserved synthetic EHR data that not only maintains high quality with heterogeneous features, missing values, and irregular measures but also achieves differential privacy with enhanced privacy-utility trade-off. Extensive experiments prove that IGAMT significantly outperforms baseline and state-of-the-art models in terms of resemblance to real data and performance of downstream applications. Ablation studies also prove the effectiveness of the techniques applied in IGAMT.

38. · 100% match · 2025

DMAP: A Deep-Model Driven Multi-Agent Framework for Reliable Diagnosis Prediction ([link](#))

Wenjing Yu et al.

2025 IEEE International Conference on Bioinformatics and Biomedicine (BIBM) · Dec 15, 2025 · 0 citations

Deep learning based diagnosis prediction from Electronic Health Record (EHR) data has demonstrated strong performance but remains hindered by limited interpretability, reducing its clinical applicability. Existing multi-agent frameworks attempt to address this challenge; however, they often rely on superficial, semantics-only predictions from a large language model (LLM) or adopt multi-classification heads, thereby inheriting the opacity of conventional deep learning approaches. We introduce DMAP, a multi-agent framework designed to enhance EHR modeling through structured collaboration and interpretable decision-making. Inspired by multidisciplinary clinical workflows, DMAP employs three types of agents: DeepL Agents, a Leader Agent, and a Critical Agent, which work together to

analyze patient data and generate reliable reports. DeepL Agents process structured EHR inputs, model outputs, and interpretability signals to provide initial clinical evaluations. Leader Agent coordinates these insights through iterative discussions, synthesizing consensus-driven disease predictions and draft report. Critical Agent evaluates the medical soundness of the draft report, identifying gaps or inconsistencies. When uncertainty arises, a retrieval-augmented generation (RAG) module integrates medical knowledge to support final decisions. By simulating expert collaboration and integrating structured reasoning with external knowledge, DMAP delivers reliable, interpretable predictions and reports. Extensive experiments on two EHR datasets demonstrate its superior performance in disease prediction and report generation, highlighting its potential to advance clinical decision support through explainable and adaptive multi-agent collaboration.

39. · 100% match · 2025 · 2.4 cit/yr

Advancing Continuous Prediction for Acute Kidney Injury via Multi-Task Learning: Towards Better Clinical Applicability ([link](#))

Hyunwoo Kim et al.

IEEE Journal of Biomedical and Health Informatics · Apr 10, 2025 · 3 citations

Acute kidney injury (AKI) presents a public health challenge with profound short and long-term morbidity and mortality. Early prediction and severity identification of AKI are crucial for improving clinical outcomes through timely interventions and efficient resource allocation. Previous studies have predominantly focused on serum creatinine, neglecting the significance of urine output, which, combined with the delayed rise in serum creatinine post-AKI onset, hinders the timely detection of AKI. To address these shortcomings, we propose a novel multi-task learning approach incorporating a continuous urine output monitoring strategy, predicting AKI onset and stage within 6-hour intervals up to 48 hours. Our model exhibits strong performance with area under the receiver operating characteristic curve (ROC) 99.3% and area under the precision-recall curve (PRC) 99.0% for predicting AKI within 48 hours, ROC 97.4% and PRC 98.6% for predicting renal replacement therapy for AKI patients. Also, our model is able to capture overall disease trends perfectly for 35.7% of the AKI cohort and 94.8% of the disease-free cohort. The proposed approach enhances clinical applicability, providing insights into disease dynamics.

40. · 100% match · 2026

ClinicalAgents: Multi-Agent Orchestration for Clinical Decision Making with Dual-Memory ([link](#))

Zhuohan Ge et al.

Mar 27, 2026 · 0 citations

While Large Language Models (LLMs) have demonstrated potential in healthcare, they often struggle with the complex, non-linear reasoning required for accurate clinical diagnosis. Existing methods typically rely on static, linear mappings from symptoms to diagnoses, failing to capture the iterative, hypothesis-driven reasoning inherent in human clinicians. To bridge this gap, we introduce ClinicalAgents, a novel multi-agent framework designed to simulate the cognitive workflow of expert clinicians.

Unlike rigid sequential chains, ClinicalAgents employs a dynamic orchestration mechanism modeled as a Monte Carlo Tree Search (MCTS) process. This allows an orchestrator to iteratively generate hypotheses, actively verify evidence, and trigger backtracking when critical information is missing. The foundation of this framework is a Dual-Memory architecture: a mutable working memory that maintains the evolving patient state for context-aware reasoning, and a static experience memory that retrieves clinical guidelines and historical cases via an active feedback loop. Extensive experiments demonstrate that ClinicalAgents achieves the best performance among evaluated baselines, significantly enhancing both diagnostic accuracy and explainability compared to strong single-agent and multi-agent baselines. Our code is released at <https://github.com/ZhuohanGe/ClinicalAgents-Code>.

41. · 100% match · 2017 · 16 cit/yr

Prediction of Acute Kidney Injury With a Machine Learning Algorithm Using Electronic Health Record Data ([link](#))

Hamid Mohamadlou et al.

Canadian Journal of Kidney Health and Disease · Nov 22, 2017 · 139 citations

Background: A major problem in treating acute kidney injury (AKI) is that clinical criteria for recognition are markers of established kidney damage or impaired function; treatment before such damage manifests is desirable. Clinicians could intervene during what may be a crucial stage for preventing permanent kidney injury if patients with incipient AKI and those at high risk of developing AKI could be identified. Objective: In this study, we evaluate a machine learning algorithm for early detection and prediction of AKI. Design: We used a machine learning technique, boosted ensembles of decision trees, to train an AKI prediction tool on retrospective data taken from more than 300 000 inpatient encounters. Setting: Data were collected from inpatient wards at Stanford Medical Center and intensive care unit patients at Beth Israel Deaconess Medical Center. Patients: Patients older than the age of 18 whose hospital stays lasted between 5 and 1000 hours and who had at least one documented measurement of heart rate, respiratory rate, temperature, serum creatinine (SCr), and Glasgow Coma Scale (GCS). Measurements: We tested the algorithm's ability to detect AKI at onset and to predict AKI 12, 24, 48, and 72 hours before onset. Methods: We tested AKI detection and prediction using the National Health Service (NHS) England AKI Algorithm as a gold standard. We additionally tested the algorithm's ability to detect AKI as defined by the Kidney Disease: Improving Global Outcomes (KDIGO) guidelines. We compared the algorithm's 3-fold cross-validation performance to the Sequential Organ Failure Assessment (SOFA) score for AKI identification in terms of area under the receiver operating characteristic (AUROC). Results: The algorithm demonstrated high AUROC for detecting and predicting NHS-defined AKI at all tested time points. The algorithm achieves AUROC of 0.872 (95% confidence interval [CI], 0.867-0.878) for AKI detection at time of onset. For prediction 12 hours before onset, the algorithm achieves an AUROC of 0.800 (95% CI, 0.792-0.809). For 24-hour predictions, the algorithm achieves AUROC of 0.795 (95% CI, 0.785-0.804). For 48-hour and 72-hour predictions, the algorithm achieves AUROC values of 0.761 (95% CI, 0.753-0.768) and 0.728 (95% CI, 0.719-0.737), respectively. Limitations: Because of the retrospective nature of this study, we cannot draw any conclusions about the impact the algorithm's predictions will have on patient outcomes in a clinical setting. Conclusions: The results of these experiments suggest that a machine learning-based AKI prediction tool may offer important

prognostic capabilities for determining which patients are likely to suffer AKI, potentially allowing clinicians to intervene before kidney damage manifests.

42. · 100% match · 2023 · 1.5 cit/yr

Redefining digital health interfaces with large language models ([link](#))

F. Imrie, Paulius Rauba, and M. Schaar

Frontiers in Artificial Intelligence · Oct 5, 2023 · 4 citations

Digital health tools have the potential to significantly improve the delivery of healthcare services. However, their adoption remains comparatively limited due, in part, to challenges surrounding usability and trust. Large Language Models (LLMs) have emerged as general-purpose models with the ability to process complex information and produce human-quality text, presenting a wealth of potential applications in healthcare. Directly applying LLMs in clinical settings is not straightforward, however, as LLMs are susceptible to providing inconsistent or nonsensical answers. We demonstrate how LLM-based systems, with LLMs acting as agents, can utilize external tools and provide a novel interface between clinicians and digital technologies. This enhances the utility and practical impact of digital healthcare tools and AI models while addressing current issues with using LLMs in clinical settings, such as hallucinations. We illustrate LLM-based interfaces with examples of cardiovascular disease and stroke risk prediction, quantitatively assessing their performance and highlighting the benefit compared to traditional interfaces for digital tools.

43. · 100% match · 2024 · 33 cit/yr

MedCalc-Bench: Evaluating Large Language Models for Medical Calculations ([link](#))

Nikhil Khandekar et al.

ArXiv · Jun 17, 2024 · 67 citations

As opposed to evaluating computation and logic-based reasoning, current benchmarks for evaluating large language models (LLMs) in medicine are primarily focused on question-answering involving domain knowledge and descriptive reasoning. While such qualitative capabilities are vital to medical diagnosis, in real-world scenarios, doctors frequently use clinical calculators that follow quantitative equations and rule-based reasoning paradigms for evidence-based decision support. To this end, we propose MedCalc-Bench, a first-of-its-kind dataset focused on evaluating the medical calculation capability of LLMs. MedCalc-Bench contains an evaluation set of over 1000 manually reviewed instances from 55 different medical calculation tasks. Each instance in MedCalc-Bench consists of a patient note, a question requesting to compute a specific medical value, a ground truth answer, and a step-by-step explanation showing how the answer is obtained. While our evaluation results show the potential of LLMs in this area, none of them are effective enough for clinical settings. Common issues include extracting the incorrect entities, not using the correct equation or rules for a calculation task, or incorrectly performing the arithmetic for the computation. We hope our study highlights the quantitative knowledge and reasoning gaps in LLMs within medical settings, encouraging future improvements of LLMs for various clinical calculation tasks. 1

44. · 100% match · 2026 · 6.0 cit/yr

EHR-RAG: Bridging Long-Horizon Structured Electronic Health Records and Large Language Models via Enhanced Retrieval-Augmented Generation ([link](#))

Lang Cao, Qingyu Chen, and Yue Guo

ArXiv · Jan 29, 2026 · 3 citations

Electronic Health Records (EHRs) provide rich longitudinal clinical evidence that is central to medical decision-making, motivating the use of retrieval-augmented generation (RAG) to ground large language model (LLM) predictions. However, long-horizon EHRs often exceed LLM context limits, and existing approaches commonly rely on truncation or vanilla retrieval strategies that discard clinically relevant events and temporal dependencies. To address these challenges, we propose EHR-RAG, a retrieval-augmented framework designed for accurate interpretation of long-horizon structured EHR data. EHR-RAG introduces three components tailored to longitudinal clinical prediction tasks: Event- and Time-Aware Hybrid EHR Retrieval to preserve clinical structure and temporal dynamics, Adaptive Iterative Retrieval to progressively refine queries in order to expand broad evidence coverage, and Dual-Path Evidence Retrieval and Reasoning to jointly retrieves and reasons over both factual and counterfactual evidence. Experiments across four long-horizon EHR prediction tasks show that EHR-RAG consistently outperforms the strongest LLM-based baselines, achieving an average Macro-F1 improvement of 10.76%. Overall, our work highlights the potential of retrieval-augmented LLMs to advance clinical prediction on structured EHR data in practice.

45. · 100% match · 2023

A dual-channel deep learning approach to continuous prediction of acute kidney injury in the intensive care unit ([link](#))

Xuetong Kong et al.

2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM) · Dec 5, 2023 · 0 citations

Acute Kidney Injury (AKI) often occurs in the intensive care units (ICU), where it is associated with high morbidity and mortality. Early and continuous prediction of AKI plays a crucial role in preventing AKI events, providing timely treatment, and reducing mortality. Continuously monitoring the high dimensional vital signs and lab measurements of the patients has always been a challenging task. In this study, we proposed a novel dual-channel deep learning approach named DC-AKI to address this challenge. DC-AKI extracts feature of temporal variables at different granularity, comprehensively and accurately obtains effective information, and finally continuous predicting risks of the patients. One channel of the model calculates corresponding local interactions of the variables with convolution kernels. The other channel uses the gated recurrent units (GRU) network and attention mechanism to obtains global interactions information among the variables. Finally, the model uses full connection layer to combine the representation vectors of the two channels, and uses Sigmoid classifier for prediction. We trained and tested our model on the MIMIC IV dataset, and the AUC ROC value of DC-AKI for predicting the risk of AKI in ICU patients 48 hours in advance was 0.9527, outperforming other methods. And our model was externally validated on the eICU dataset with an AUC ROC of 0.9054. These results demonstrate the robustness and accuracy of the model on different datasets. This finding provides clinicians with an opportunity to identify and manage patients with AKI in a timely manner,

thereby improving patient prognosis. The implemented code is available online at <https://github.com/BioMedBigDataCenter/DC-AKI>.

46. · 100% match · 2023 · 5.7 cit/yr

Acute kidney injury associated with nephrotoxic drugs in critically ill patients: a multicenter cohort study using electronic health record data ([link](#))

Izak A R Yasrebi-de Kom et al.

Clinical Kidney Journal · Jul 5, 2023 · 17 citations

ABSTRACT Background Nephrotoxic drugs frequently cause acute kidney injury (AKI) in adult intensive care unit (ICU) patients. However, there is a lack of large pharmaco-epidemiological studies investigating the associations between drugs and AKI. Importantly, AKI risk factors may also be indications or contraindications for drugs and thereby confound the associations. Here, we aimed to estimate the associations between commonly administered (potentially) nephrotoxic drug groups and AKI in adult ICU patients whilst adjusting for confounding. Methods In this multicenter retrospective observational study, we included adult ICU admissions to 13 Dutch ICUs. We measured exposure to 44 predefined (potentially) nephrotoxic drug groups. The outcome was AKI during ICU admission. The association between each drug group and AKI was estimated using etiological cause-specific Cox proportional hazard models and adjusted for confounding. To facilitate an (independent) informed assessment of residual confounding, we manually identified drug group-specific confounders using a large drug knowledge database and existing literature. Results We included 92 616 ICU admissions, of which 13 492 developed AKI (15%). We found 14 drug groups to be associated with a higher hazard of AKI after adjustment for confounding. These groups included established (e.g. aminoglycosides), less well established (e.g. opioids) and controversial (e.g. sympathomimetics with α - and β -effect) drugs. Conclusions The results confirm existing insights and provide new ones regarding drug associated AKI in adult ICU patients. These insights warrant caution and extra monitoring when prescribing nephrotoxic drugs in the ICU and indicate which drug groups require further investigation.

47. · 100% match · 2019 · 21 cit/yr

A prospective multi-center quality improvement initiative (NINJA) indicates a reduction in nephrotoxic acute kidney injury in hospitalized children. ([link](#))

S. Goldstein et al.

Kidney international · Nov 1, 2019 · 139 citations

Nephrotoxic medication (NTMx) exposure is a common cause of acute kidney injury (AKI) in hospitalized children. The Nephrotoxic Injury Negated by Just-in time Action (NINJA) program decreased NTMx associated AKI (NTMx-AKI) by 62% at one center. To further test the program, we incorporated NINJA across nine centers with the goal of reducing NTMx exposure and, consequently, AKI rates across these centers. NINJA screens all non-critically ill hospitalized patients for high NTMx exposure (over three medications on the same day or an intravenous aminoglycoside over three consecutive days), and then recommends obtaining a daily serum creatinine level in exposed patients for the duration of, and two days after, exposure ending. Additionally, substitution of equally efficacious but less

nephrotoxic medications for exposed patients starting the day of exposure was recommended when possible. The main outcome was AKI as defined by the Kidney Disease Improving Global Outcomes (KDIGO) serum creatinine criteria (increase of 50% or 0.3 mg/dl over baseline). The primary outcome measure was AKI episodes per 1000 patient-days. Improvement was defined by statistical process control methodology and confirmed by Autoregressive Integrated Moving Average (ARIMA) modeling. Eight consecutive bi-weekly measure rates in the same direction from the established baseline qualified as special cause change for special process control. We observed a significant and sustained 23.8% decrease in NTMx-AKI rates by statistical process control analysis and by ARIMA modeling; similar to those of the pilot single center. Thus, we have successfully applied the NINJA program to multiple pediatric institutions yielding decreased AKI rates.

48. · 100% match · 2025 · 3.5 cit/yr

Development and Validation of a CatBoost-Based Model for Predicting Significant Creatinine Elevation in ICU Patients Receiving Vancomycin Therapy ([link](#))

Junyi Fan et al.

BioMedInformatics · Dec 10, 2025 · 2 citations

Vancomycin remains a cornerstone for severe Gram-positive infections in the ICU, yet creatinine elevation—a sensitive marker of early renal stress—occurs frequently and complicates therapy. We developed a machine learning model to predict vancomycin-associated creatinine elevation using routinely available clinical data, enabling preemptive risk stratification. In this retrospective MIMIC-IV cohort study (n=10,288 ICU adults aged 18–80 receiving vancomycin), the primary outcome was creatinine elevation per KDIGO criteria (≥ 0.3 mg/dL within 48 h or $\geq 50\%$ within 7 d). A two-stage feature selection (SelectKBest + Random Forest) identified 15 predictors from 30 candidates. Six algorithms were compared via 5-fold cross-validation. CatBoost was selected for final modeling; interpretability was assessed using SHAP values and Accumulated Local Effects (ALE) plots. Creatinine elevation occurred in 2903 patients (28.2%). CatBoost achieved AUROC 0.818 (95% CI: 0.801–0.834), sensitivity 0.800, specificity 0.681, and NPV 0.900. Top predictors were serum phosphate, total bilirubin, magnesium, Charlson Comorbidity Index, and APSIII score. SHAP analysis confirmed hyperphosphatemia as the strongest driver; ALE plots revealed non-linear, clinically plausible thresholds (e.g., phosphate >4.5 mg/dL sharply increased risk). This interpretable model accurately predicts vancomycin-associated creatinine elevation using standard ICU monitoring data. With high negative predictive value, it supports early exclusion of low-risk patients and targeted interventions (e.g., intensified TDM, nephrotoxin avoidance) in high-risk cases—facilitating precision antimicrobial stewardship in critical care.

49. · 100% match · 2017 · 106 cit/yr

Real-valued (Medical) Time Series Generation with Recurrent Conditional GANs ([link](#))

Cristóbal Esteban, Stephanie L. Hyland, and G. Rätsch

ArXiv · Jun 8, 2017 · 966 citations

Generative Adversarial Networks (GANs) have shown remarkable success as a framework for training models to produce realistic-looking data. In this work, we propose a Recurrent GAN (RGAN) and

Recurrent Conditional GAN (RCGAN) to produce realistic real-valued multi-dimensional time series, with an emphasis on their application to medical data. RGANs make use of recurrent neural networks (RNNs) in the generator and the discriminator. In the case of RCGANs, both of these RNNs are conditioned on auxiliary information. We demonstrate our models in a set of toy datasets, where we show visually and quantitatively (using sample likelihood and maximum mean discrepancy) that they can successfully generate realistic time-series. We also describe novel evaluation methods for GANs, where we generate a synthetic labelled training dataset, and evaluate on a real test set the performance of a model trained on the synthetic data, and vice-versa. We illustrate with these metrics that RCGANs can generate time-series data useful for supervised training, with only minor degradation in performance on real test data. This is demonstrated on digit classification from ‘serialised’ MNIST and by training an early warning system on a medical dataset of 17,000 patients from an intensive care unit. We further discuss and analyse the privacy concerns that may arise when using RCGANs to generate realistic synthetic medical time series data, and demonstrate results from differentially private training of the RCGAN.

50. · 100% match · 2023

Causal prediction models for medication safety monitoring: The diagnosis of vancomycin-induced acute kidney injury ([link](#))

Izak A. R. Yasrebi-de Kom et al.

ArXiv · Nov 15, 2023 · 0 citations

The current best practice approach for the retrospective diagnosis of adverse drug events (ADEs) in hospitalized patients relies on a full patient chart review and a formal causality assessment by multiple medical experts. This evaluation serves to qualitatively estimate the probability of causation (PC); the probability that a drug was a necessary cause of an adverse event. This practice is manual, resource intensive and prone to human biases, and may thus benefit from data-driven decision support. Here, we pioneer a causal modeling approach using observational data to estimate a lower bound of the PC (PC_{low}). This method includes two key causal inference components: (1) the target trial emulation framework and (2) estimation of individualized treatment effects using machine learning. We apply our method to the clinically relevant use-case of vancomycin-induced acute kidney injury in intensive care patients, and compare our causal model-based PC_{low} estimates to qualitative estimates of the PC provided by a medical expert. Important limitations and potential improvements are discussed, and we conclude that future improved causal models could provide essential data-driven support for medication safety monitoring in hospitalized patients.

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